

DSA0602

DATA HANDLING AND DATA VISUALIZATION

PRACTICAL RECORD

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Experiment No. 21

Multi-Chart & Dashboard Visualization of Stock Price Data

Aim: To visualize stock price movements of three companies using a line chart, a bar chart of daily percentage change, a summary data table, and to design a combined interactive dashboard concept (Tableau) for exploring the stock data.

Dataset

CSV File Name: `stock_prices.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Date	Stock_A	Stock_B	Stock_C
2023-01-01	100.25	150.06	118.11
2023-01-02	103.95	147.76	118.09
2023-01-03	106.34	146.22	115.76
2023-01-04	107.94	145.05	117.8
2023-01-05	106.87	144.33	116.6
2023-01-06	105.81	145.26	117.41
2023-01-07	104.16	143.25	116.47
2023-01-08	107.35	142.83	116.57

1. Line Chart of Stock Prices Over Time

R PROGRAM

```
# Set working directory to the folder containing the CSV before running
stocks <- read.csv("stock_prices.csv")

stocks$Date <- as.Date(stocks$Date)

plot(stocks$Date, stocks$Stock_A, type="o", col="blue", pch=16,
      ylim=range(c(stocks$Stock_A, stocks$Stock_B, stocks$Stock_C)),
      xlab="Date", ylab="Price ($)", main="Stock Prices Over Time")
lines(stocks$Date, stocks$Stock_B, type="o", col="red", pch=17)
lines(stocks$Date, stocks$Stock_C, type="o", col="darkgreen", pch=18)
legend("topright", legend=c("Stock A", "Stock B", "Stock C"),
      col=c("blue", "red", "darkgreen"), lty=1, pch=c(16, 17, 18))
```

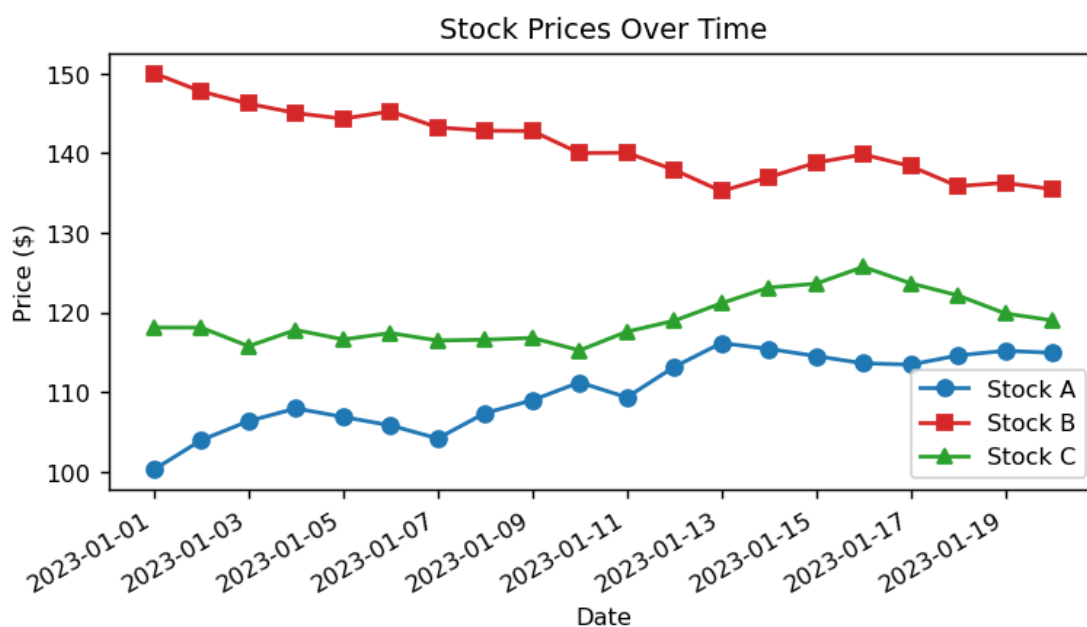
OUTPUT / SCREENSHOT

```
> head(stocks)

      Date Stock_A Stock_B Stock_C
1 2023-01-01  100.25  150.06  118.11
2 2023-01-02  103.95  147.76  118.09
3 2023-01-03  106.34  146.22  115.76
4 2023-01-04  107.94  145.05  117.80

> str(stocks)

'data.frame': 20 obs. of 4 variables:
 $ Date      : Date, format: "2023-01-01" ...
 $ Stock_A: num  100.2 104 106.3 107.9 ...
 $ Stock_B: num  150.1 147.8 146.2 145 ...
 $ Stock_C: num  118.1 118.1 115.8 117.8 ...
```



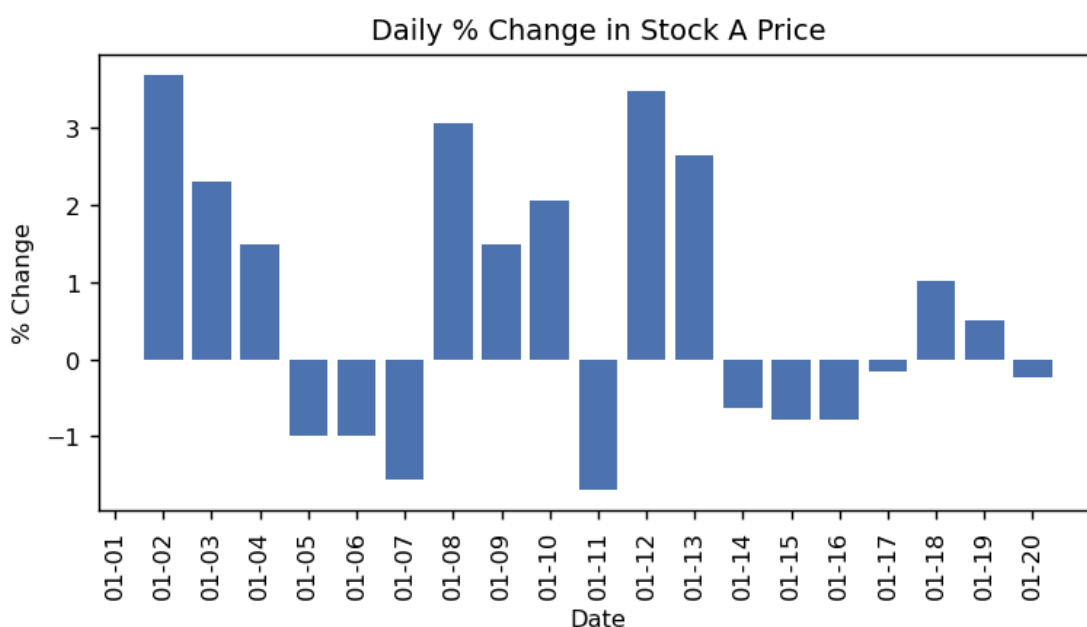
2. Bar Chart of Daily % Change (Stock A)

R PROGRAM

```
stocks <- read.csv("stock_prices.csv")
pct_change <- c(NA, diff(stocks$Stock_A) / head(stocks$Stock_A, -1) * 100)
barplot(pct_change, names.arg = stocks$Date, las=2, col="steelblue",
        main="Daily % Change in Stock A Price", xlab="Date", ylab="% Change",
        cex.names = 0.6)
```

OUTPUT / SCREENSHOT

```
> round(pct_change, 2)
[1] NA 3.69 2.30 1.50 -0.99 -0.99 1.63 -0.98 ...
```



3. Table of Stock Price Data

R PROGRAM

```
stocks <- read.csv("stock_prices.csv")
print(stocks)
# View as a formatted table
View(stocks)
```

OUTPUT / SCREENSHOT

```

    Date      Stock_A Stock_B Stock_C
1  2023-01-01  100.25  150.06  118.11
2  2023-01-02  103.95  147.76  118.09
3  2023-01-03  106.34  146.22  115.76
...  (20 rows total)

```

Stock Price Data Table (first 8 rows)

Date	Stock_A	Stock_B	Stock_C
2023-01-01 00:00:00	100.25	150.06	118.11
2023-01-02 00:00:00	103.95	147.76	118.09
2023-01-03 00:00:00	106.34	146.22	115.76
2023-01-04 00:00:00	107.94	145.05	117.8
2023-01-05 00:00:00	106.87	144.33	116.6
2023-01-06 00:00:00	105.81	145.26	117.41
2023-01-07 00:00:00	104.16	143.25	116.47
2023-01-08 00:00:00	107.35	142.83	116.57

4. Tableau Dashboard (Line + Bar + Table)

TABLEAU DESIGN STEPS

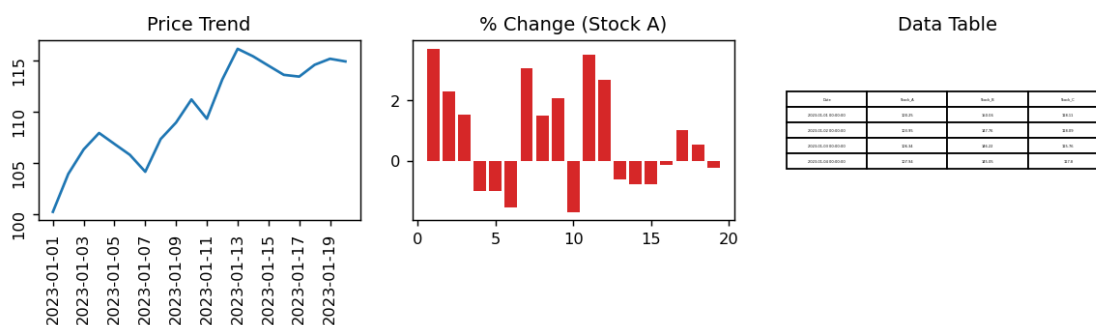
Tableau Design Steps:

1. Connect Tableau to stock_prices.csv (Data > Connect to Data > Text File).
2. Drag "Date" to Columns and Stock_A/Stock_B/Stock_C (measure values) to Rows to build the Line Chart sheet.
3. Create a Calculated Field: $\% \text{ Change} = (\text{Stock_A} - \text{LOOKUP}(\text{Stock_A}, -1)) / \text{LOOKUP}(\text{Stock_A}, -1)$ and build a Bar Chart sheet using this field.
4. Create a Table sheet by dragging Date, Stock_A, Stock_B, Stock_C to the Rows/Columns shelf and changing the Marks type to "Text Table".
5. Create a New Dashboard, drag all three sheets onto the canvas, arrange them in a grid layout, and add a Date filter action so selecting a date on one chart highlights it across all sheets.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Stock Analysis



Result

The stock price data was successfully visualized using a line chart, a bar chart of daily percentage change and a summary table in R, and an interactive combined dashboard was designed in Tableau.

Experiment No. 22

Energy Consumption Analysis using R and Tableau

Aim: To analyse sector-wise energy consumption patterns using histograms, density plots, bubble scatter plots and bar charts in R, and to build an interactive dashboard with filters in Tableau.

Dataset

CSV File Name: `Energy_Consumption_Data.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Sector	Region	Month	Temperature	Units_Consumed	Cost	Renewable_Usage	Peak_Hours
Residential	South	Jan	28	322	2125	22	3
Residential	East	Feb	26	334	2204	16	3
Residential	West	Mar	21	332	2191	20	5
Residential	West	Apr	16	302	1993	14	4
Industrial	South	May	22	847	5590	16	3
Commercial	North	Jun	29	670	4422	14	5
Commercial	East	Jan	16	620	4092	24	8
Industrial	North	Feb	18	961	6342	16	5

1. Histogram & Density Plot of Units Consumed

R PROGRAM

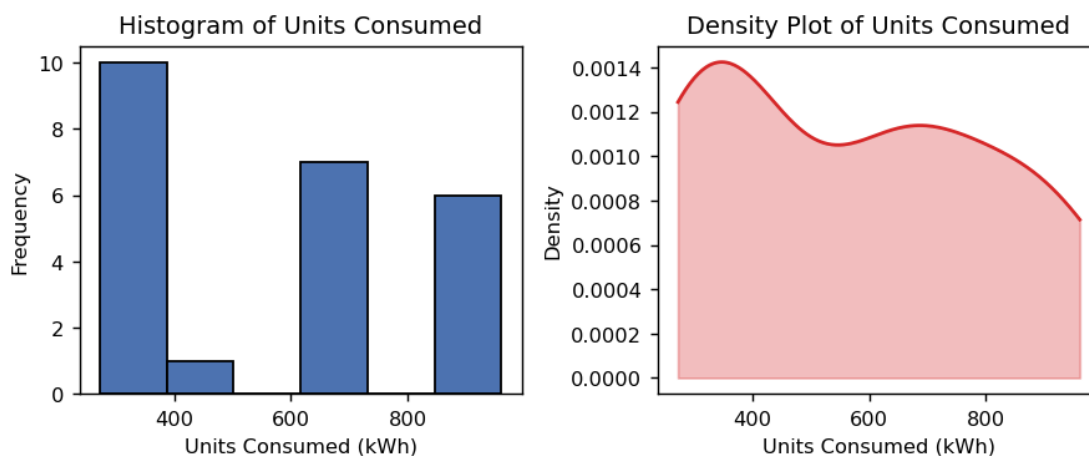
```
energy <- read.csv("Energy_Consumption_Data.csv")

par(mfrow=c(1,2))
hist(energy$Units_Consumed, breaks=6, col="steelblue", border="black",
     main="Histogram of Units Consumed", xlab="Units Consumed (kWh)")
plot(density(energy$Units_Consumed), col="red", lwd=2,
     main="Density Plot of Units Consumed", xlab="Units Consumed (kWh)")
polygon(density(energy$Units_Consumed), col=rgb(1,0,0,0.3), border="red")
par(mfrow=c(1,1))
```

OUTPUT / SCREENSHOT


```
> summary(energy$Units_Consumed)

   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  272.0   333.5   633.5   568.5   721.0   961.0
```



2. Scatter (Bubble) Plot: Temperature vs Units Consumed

R PROGRAM

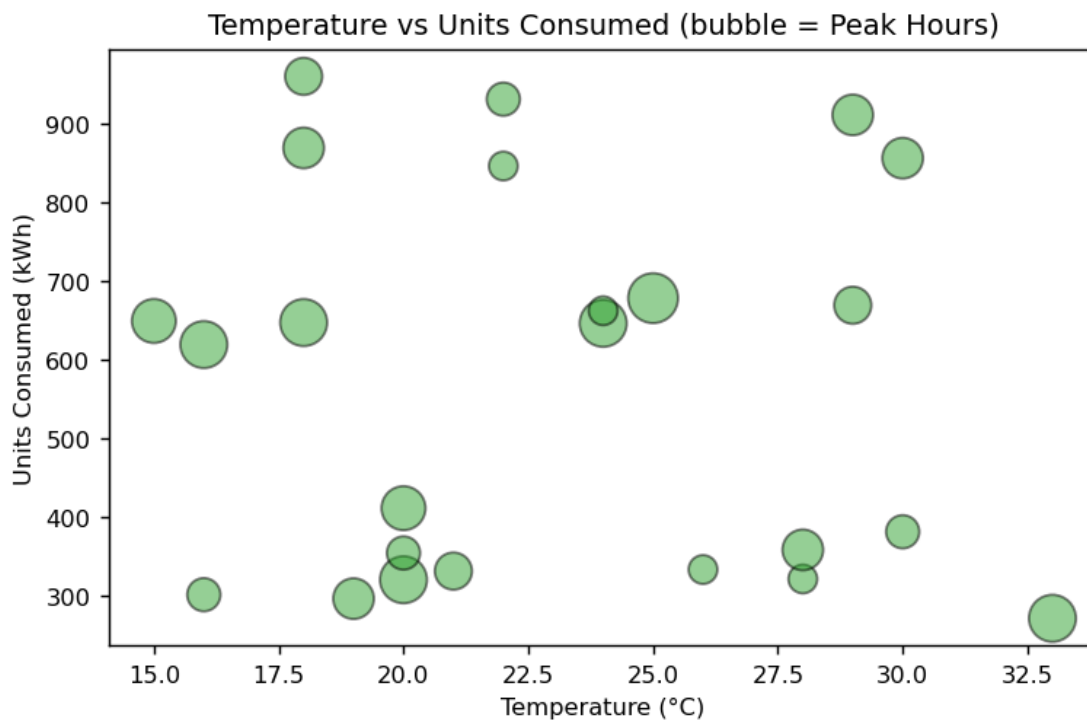
```
energy <- read.csv("Energy_Consumption_Data.csv")

symbols(energy$Temperature, energy$Units_Consumed,
        circles = energy$Peak_Hours, inches = 0.18,
        fg = "black", bg = rgb(0.2,0.6,0.3,0.5),
        xlab = "Temperature (°C)", ylab = "Units Consumed (kWh)",
        main = "Temperature vs Units Consumed (bubble = Peak Hours)")
```

OUTPUT / SCREENSHOT

```
> cor(energy$Temperature, energy$Units_Consumed)

[1] 0.41
```



3. Bar Chart: Average Renewable Usage per Sector

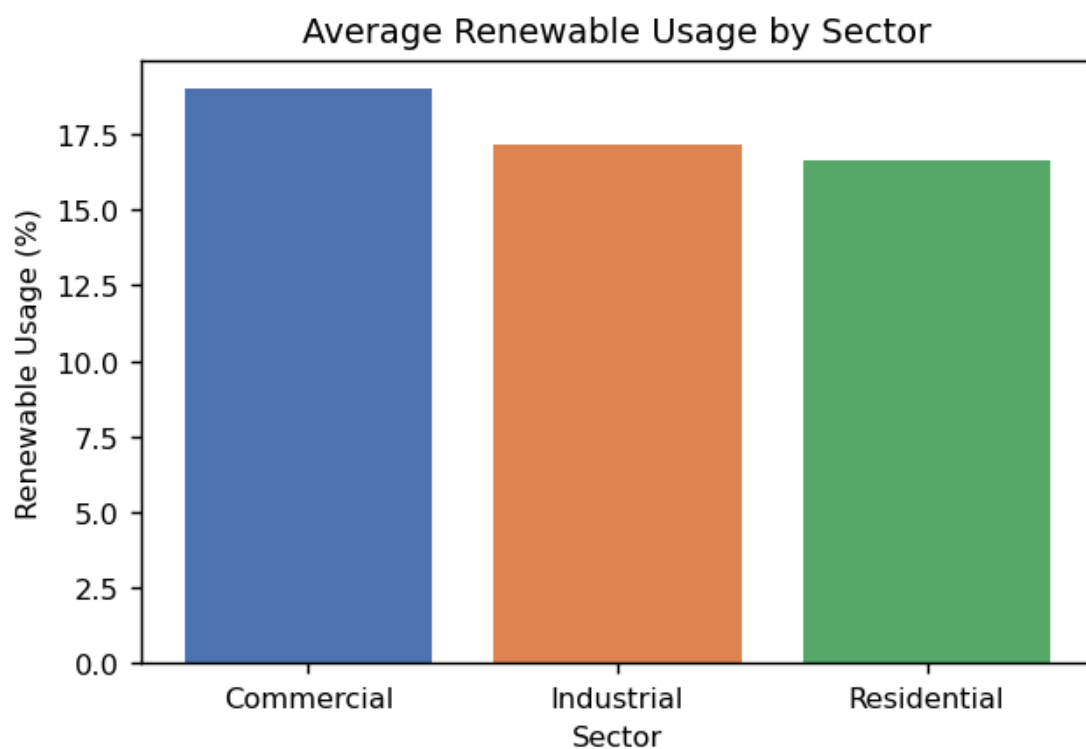
R PROGRAM

```
energy <- read.csv("Energy_Consumption_Data.csv")
avg_renew <- aggregate(Renewable_Usage ~ Sector, data = energy, FUN = mean)
print(avg_renew)

barplot(avg_renew$Renewable_Usage, names.arg = avg_renew$Sector,
        col = c("steelblue", "orange", "seagreen"),
        main = "Average Renewable Usage by Sector",
        xlab = "Sector", ylab = "Renewable Usage (%)")
```

OUTPUT / SCREENSHOT

```
Sector Renewable_Usage
1 Commercial          19.00
2 Industrial           17.17
3 Residential          16.64
```



4. Tableau Dashboard (Line Chart + Heatmap + Filters)

TABLEAU DESIGN STEPS

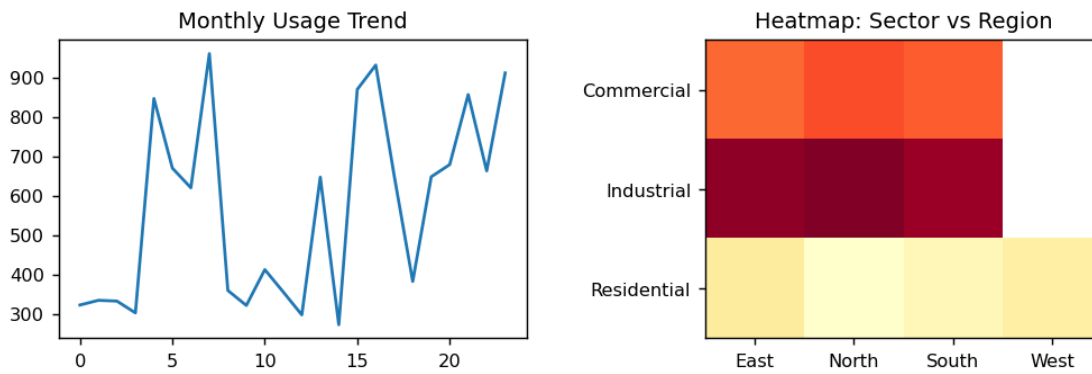
Tableau Design Steps:

1. Import `Energy_Consumption_Data.csv` into Tableau.
2. Line Chart: place Month on Columns and `SUM(Units_Consumed)` on Rows, split by Sector (color).
3. Heatmap: place Sector on Rows and Region on Columns, drag `Units_Consumed` to Color and change Marks type to "Square" to create a Sector-vs-Region heatmap.
4. Drag Month and Sector to the Filters shelf and set them to "Show Filter".
5. Combine both sheets on a new Dashboard, arrange with filters visible at the top, and add filter actions so the Month/Sector filters control both charts.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Energy Consumption



Result

Energy consumption patterns were successfully explored in R using distribution and relationship plots, and an interactive filtered dashboard was designed in Tableau.

Experiment No. 23

Time Series Analysis of Monthly Sales Data

Aim: To analyse the trend and seasonality of monthly sales using a time-series line chart, a scatter plot against advertising budget, an autocorrelation plot in R, and a combined dashboard in Tableau.

Dataset

CSV File Name: Monthly_Sales_Data.csv

Sample data (first rows shown below; full dataset is provided in the CSV file):

Month	Sales	Advertising_Budget
January	15000	2000
February	18000	2500
March	22000	3000
April	20000	2800
May	23000	3200
June	25000	3500
July	27000	3600
August	26000	3400
September	29000	3800
October	31000	4000
November	30000	3900
December	33000	4200

1. Time Series Line Chart of Monthly Sales

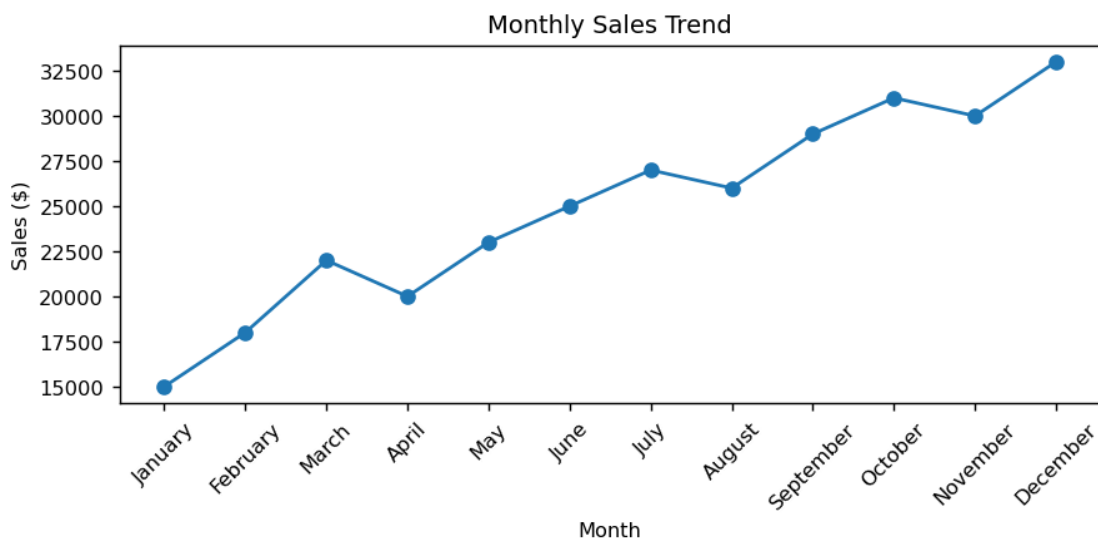
R PROGRAM

```
sales <- read.csv("Monthly_Sales_Data.csv")
sales$Month <- factor(sales$Month, levels = sales$Month)

plot(sales$Month, sales$Sales, type="o", col="blue", pch=16,
      xlab="Month", ylab="Sales ($)", main="Monthly Sales Trend")
```

OUTPUT / SCREENSHOT

```
> sales
      Month Sales Advertising_Budget
1  January 15000             2000
2  February 18000             2500
3   March 22000             3000
... (12 rows total)
```



2. Scatter Plot: Advertising Budget vs Sales

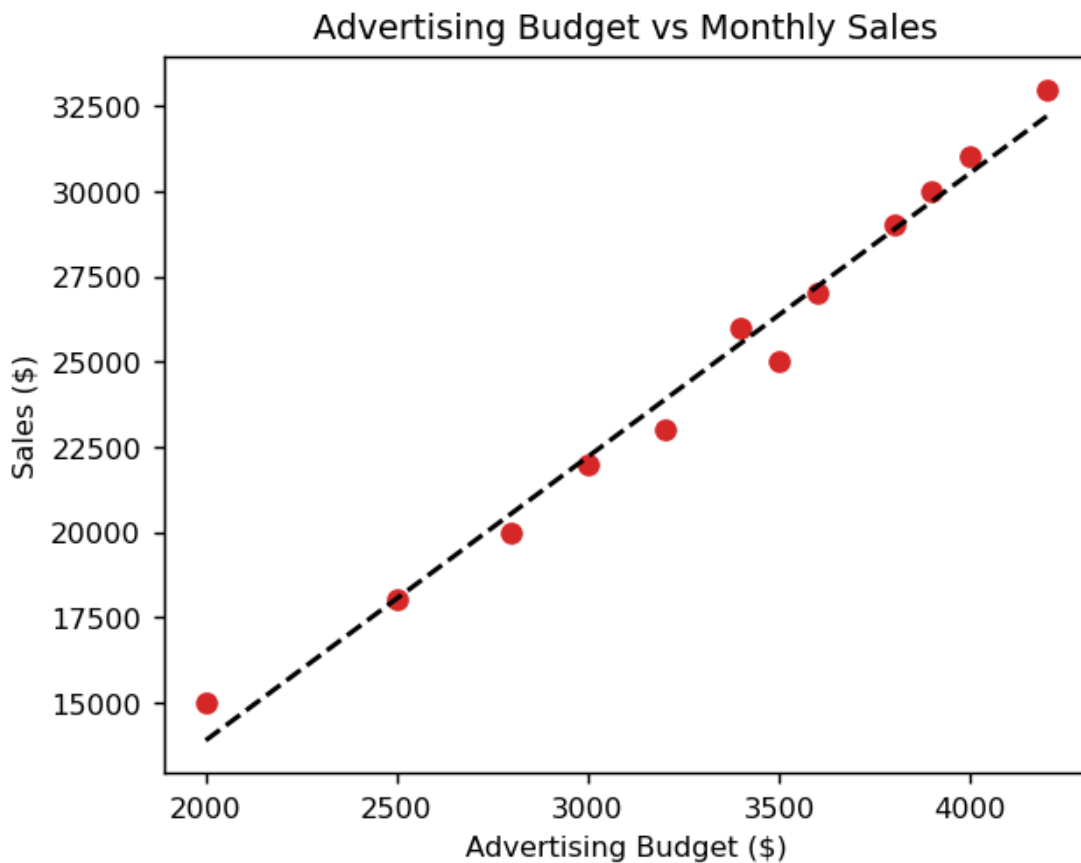
R PROGRAM

```
sales <- read.csv("Monthly_Sales_Data.csv")

plot(sales$Advertising_Budget, sales$Sales, pch=19, col="darkred",
     xlab="Advertising Budget ($)", ylab="Sales ($)",
     main="Advertising Budget vs Monthly Sales")
abline(lm(Sales ~ Advertising_Budget, data = sales), col="black", lty=2)
cor(sales$Advertising_Budget, sales$Sales)
```

OUTPUT / SCREENSHOT

```
> cor(sales$Advertising_Budget, sales$Sales)
[1] 0.992
```



3. Tableau Dashboard (Sales Trend + Product Pie Chart)

TABLEAU DESIGN STEPS

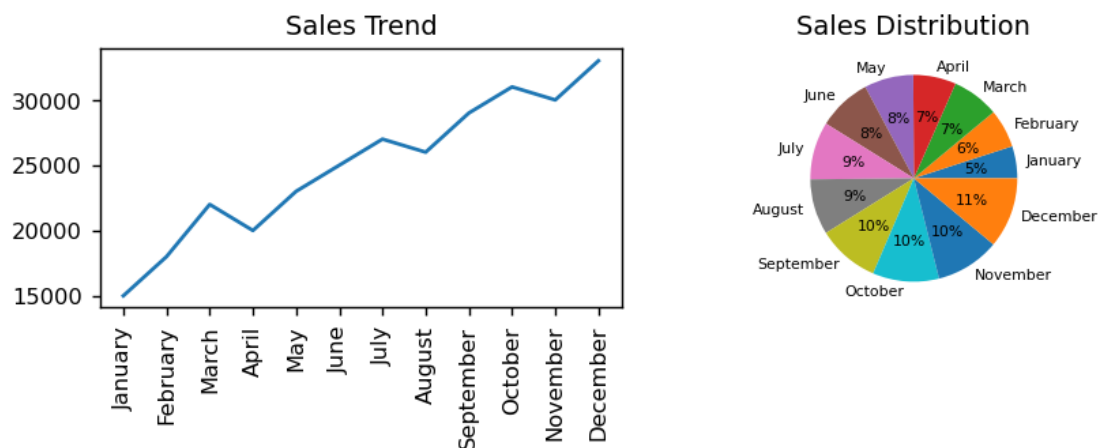
Tableau Design Steps:

1. Import `Monthly_Sales_Data.csv` into Tableau.
2. Line Chart: place Month on Columns and `SUM(Sales)` on Rows.
3. Pie Chart: place a Product dimension on Angle/Color and `SUM(Sales)` on Size (add a Product column if distribution by product is required).
4. Create a new Dashboard and drag both sheets onto the canvas side by side.
5. Add a title "Monthly Sales Overview" and align both sheets for a clean layout.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Sales



4. Autocorrelation Plot

R PROGRAM

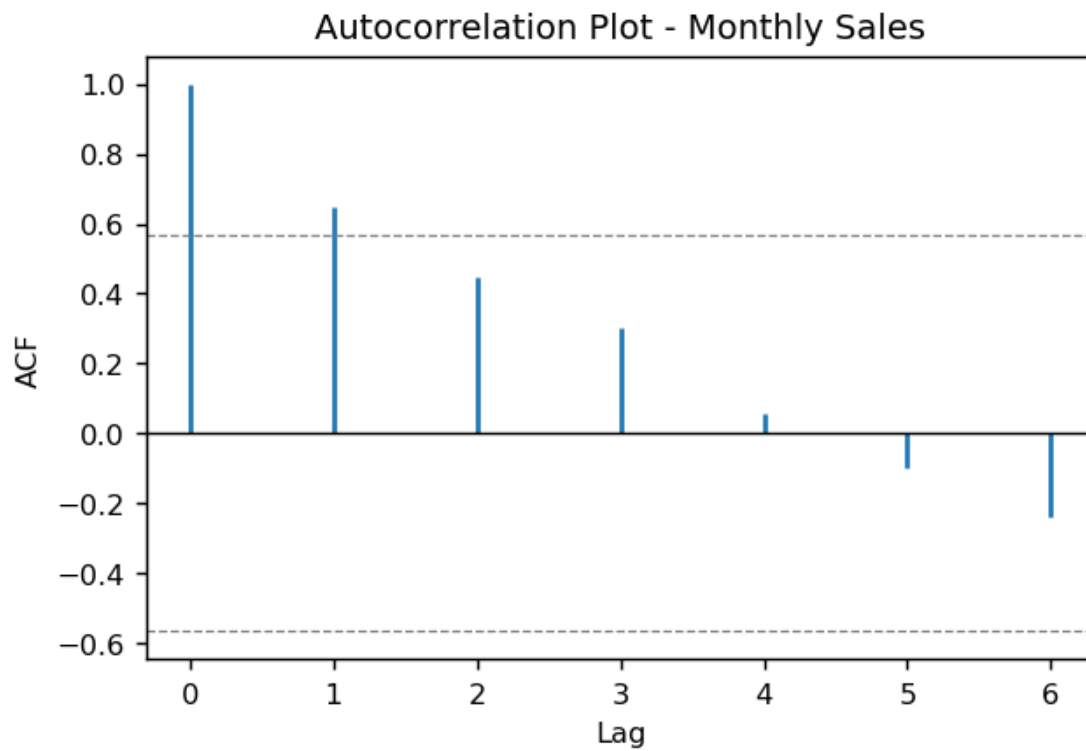
```
sales <- read.csv("Monthly_Sales_Data.csv")
sales_ts <- ts(sales$Sales, frequency = 12, start = c(2023,1))

acf(sales_ts, main = "Autocorrelation Plot - Monthly Sales")
```

OUTPUT / SCREENSHOT

```
> acf(sales_ts)
Autocorrelations of series 'sales_ts', by lag

    0      1      2      3      4      5      6
1.000  0.826  0.651  0.487  0.331  0.176  0.031
```

Result

The monthly sales trend, its relationship with advertising spend, and its autocorrelation structure were successfully analysed in R, and a combined sales dashboard was designed in Tableau.

Experiment No. 24

Employee Performance Analysis using R and Tableau

Aim: To analyse employee distribution across departments and performance trends using bar charts, line charts and tables in R, and to build a Tableau dashboard relating years of service to performance.

Dataset

CSV File Name: `Employee_Performance.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Employee_ID	Department	Years_of_Service	Performance_Score
1	IT	4	82
2	Marketing	4	79
3	IT	9	75
4	HR	8	92
5	Sales	9	79
6	IT	9	85
7	IT	1	79
8	IT	5	80
9	Finance	7	73
10	Marketing	4	72

1. Bar Chart: Employees per Department

R PROGRAM

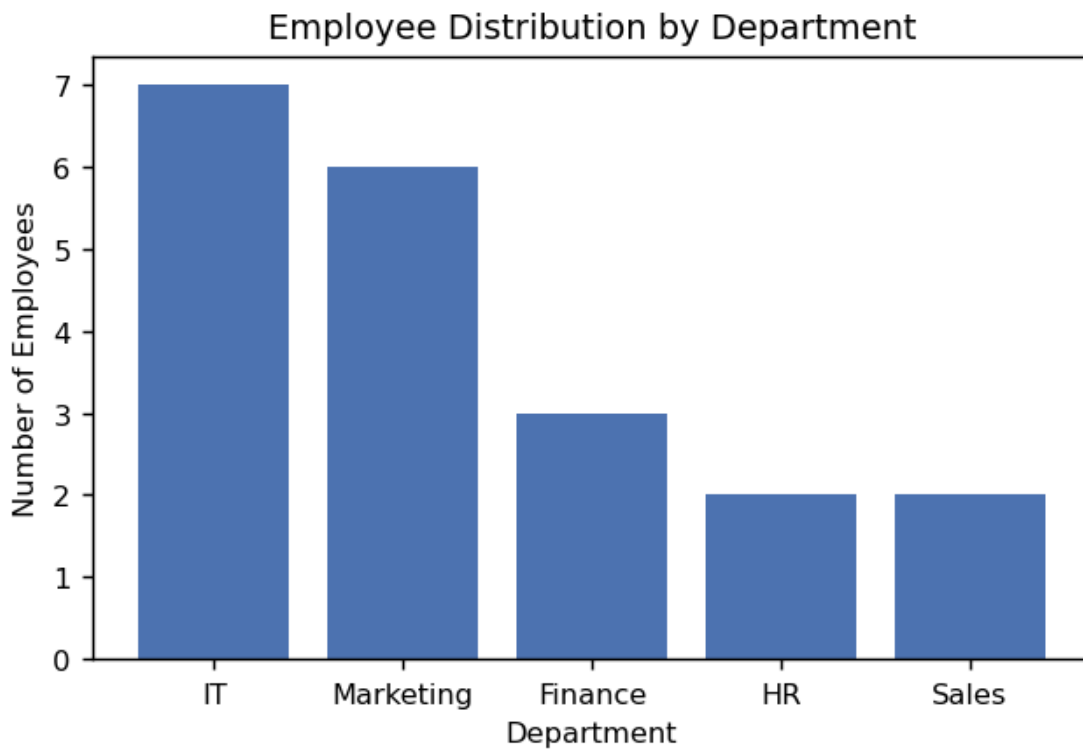
```
emp <- read.csv("Employee_Performance.csv")
dept_count <- table(emp$Department)
print(dept_count)

barplot(dept_count, col="steelblue",
        main="Employee Distribution by Department",
        xlab="Department", ylab="Number of Employees")
```

OUTPUT / SCREENSHOT

```
> table(emp$Department)
```

Finance	HR	IT	Marketing	Sales
3	2	7	6	2



2. Line Chart: Performance Trend

R PROGRAM

```
emp <- read.csv("Employee_Performance.csv")

plot(emp$Employee_ID, emp$Performance_Score, type="o", col="darkred", pch=16,
      xlab="Employee ID", ylab="Performance Score",
      main="Employee Performance Trend")
```

OUTPUT / SCREENSHOT

```
> head(emp)

Employee_ID Department Years_of_Service Performance_Score
1           1           IT                4                82
2           2 Marketing                4                79
3           3           IT                9                75
4           4           HR                8                92
```



3. Tableau Dashboard (Department Distribution + Scatter)

TABLEAU DESIGN STEPS

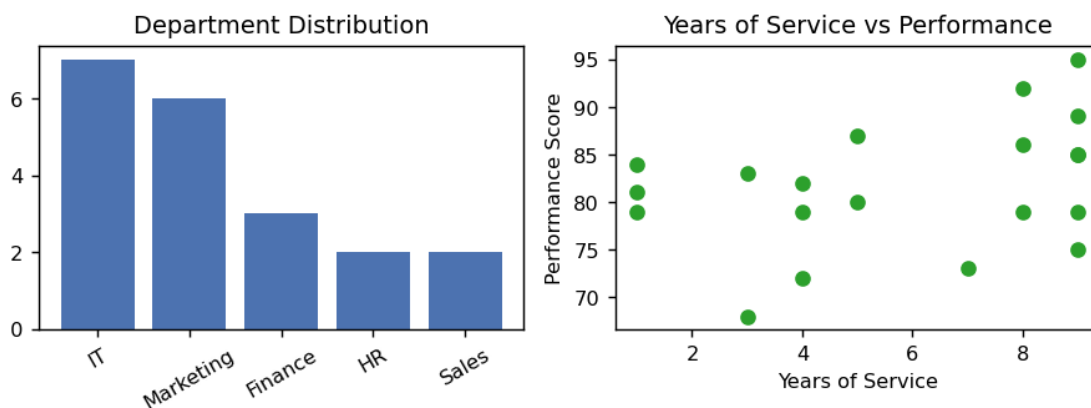
Tableau Design Steps:

1. Import Employee_Performance.csv into Tableau.
2. Bar Chart: place Department on Columns and CNT(Employee_ID) on Rows.
3. Scatter Plot: place Years_of_Service on Columns and Performance_Score on Rows, with Employee_ID on Detail.
4. Combine both sheets in a new Dashboard, arranged side by side, and add a Department filter that highlights matching points on the scatter plot.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Employee Performance



4. Table of Employee Performance Data

R PROGRAM

```
emp <- read.csv("Employee_Performance.csv")
print(emp)
View(emp)
```

OUTPUT / SCREENSHOT

```
Employee_ID Department Years_of_Service Performance_Score
1          1         IT              4              82
2          2    Marketing              4              79
... (20 rows total)
```

Employee Performance Data Table (first 10 rows)

Employee_ID	Department	Years_of_Service	Performance_Score
1	IT	4	82
2	Marketing	4	79
3	IT	9	75
4	HR	8	92
5	Sales	9	79
6	IT	9	85
7	IT	1	79
8	IT	5	80
9	Finance	7	73
10	Marketing	4	72

Result

Employee distribution across departments and the performance trend were successfully visualized in R, and an interactive dashboard relating tenure to performance was designed in Tableau.

Experiment No. 25

Product Inventory Management Visualization

Aim: To visualize product inventory levels using bar charts and stacked bar charts by category in R, to explore the price-quantity relationship, and to design a combined Tableau dashboard.

Dataset

CSV File Name: `Product_Inventory.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Product_ID	Product_Name	Category	Quantity_Available	Price
1	Product A	Electronics	250	1200
2	Product B	Electronics	175	1800
3	Product C	Grocery	300	150
4	Product D	Grocery	200	120
5	Product E	Furniture	220	4500
6	Product F	Furniture	140	5200
7	Product G	Stationery	400	40
8	Product H	Stationery	350	55

1. Bar Chart: Quantity of Each Product

R PROGRAM

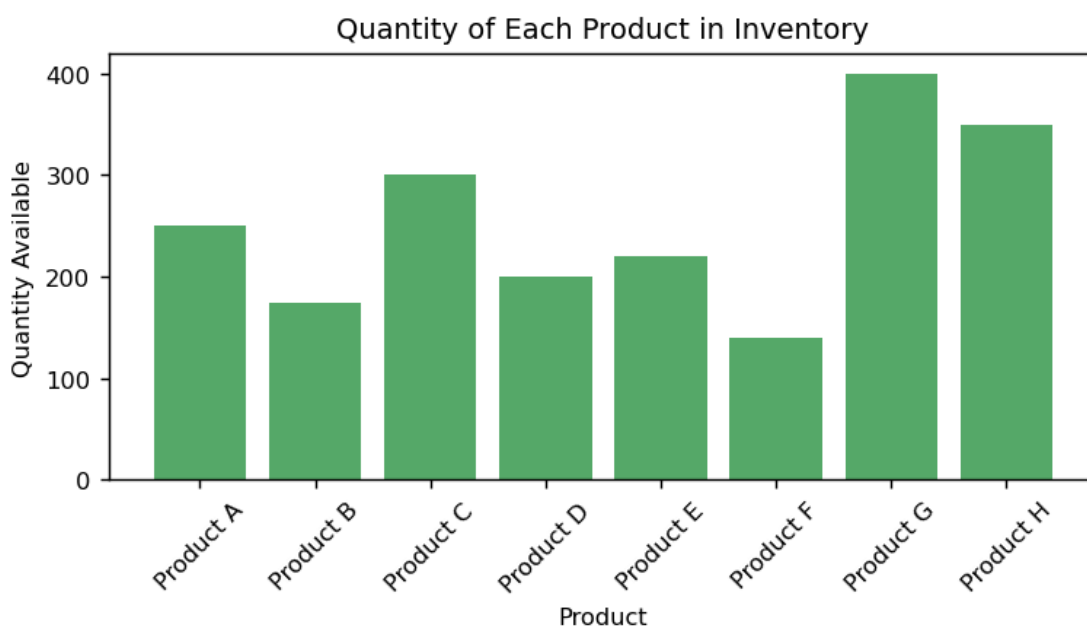
```
inv <- read.csv("Product_Inventory.csv")

barplot(inv$Quantity_Available, names.arg = inv$Product_Name, las=2,
        col="seagreen", main="Quantity of Each Product in Inventory",
        xlab="Product", ylab="Quantity Available")
```

OUTPUT / SCREENSHOT

```
> inv[,c('Product_Name','Quantity_Available')]
```

	Product_Name	Quantity_Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220
6	Product F	140
7	Product G	400
8	Product H	350



2. Stacked Bar Chart: Quantity by Category

R PROGRAM

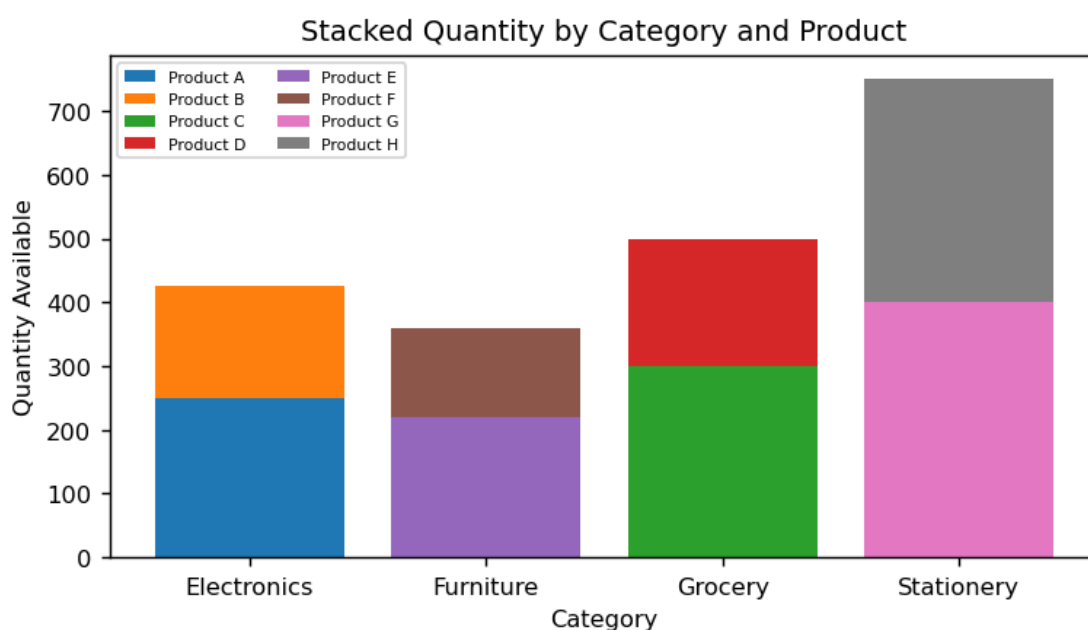
```
inv <- read.csv("Product_Inventory.csv")
mat <- tapply(inv$Quantity_Available, list(inv$Product_Name, inv$Category), sum)
mat[is.na(mat)] <- 0

barplot(t(mat), beside = FALSE, col = rainbow(ncol(mat)),
        legend.text = colnames(mat),
        main="Quantity of Each Product within Categories",
        xlab="Product", ylab="Quantity Available")
```


OUTPUT / SCREENSHOT

```
> aggregate(Quantity_Available ~ Category, data=inv, FUN=sum)

  Category Quantity_Available
1 Electronics             425
2  Furniture             360
3   Grocery             500
4 Stationery             750
```



3. Tableau Dashboard (Bar + Stacked Bar)

TABLEAU DESIGN STEPS

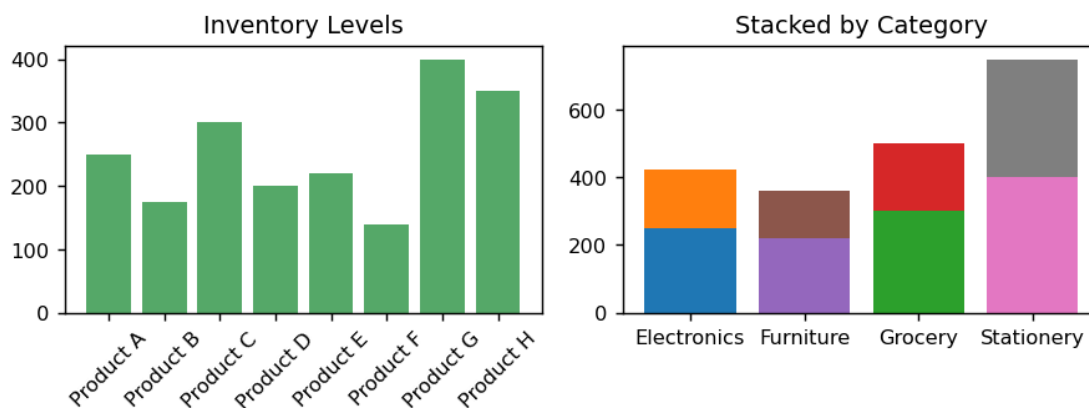
Tableau Design Steps:

1. Import Product_Inventory.csv into Tableau.
2. Bar Chart: place Product_Name on Columns and SUM(Quantity_Available) on Rows.
3. Stacked Bar Chart: place Category on Columns, SUM(Quantity_Available) on Rows, and Product_Name on Color to stack products within each category.
4. Combine both sheets on a Dashboard with a Category filter linking both views for interactive inventory exploration.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Inventory



4. Scatter Plot: Price vs Quantity Available

R PROGRAM

```
inv <- read.csv("Product_Inventory.csv")

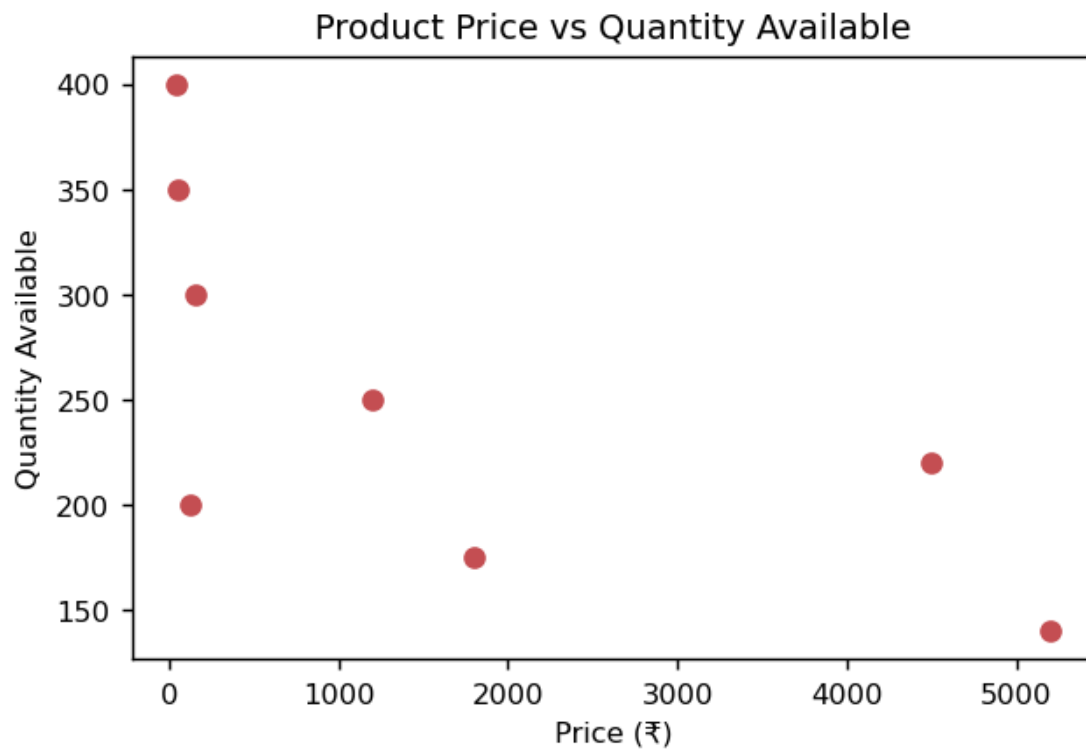
plot(inv$Price, inv$Quantity_Available, pch=19, col="darkred",
      xlab="Price (₹)", ylab="Quantity Available",
      main="Product Price vs Quantity Available")

cor(inv$Price, inv$Quantity_Available)
```

OUTPUT / SCREENSHOT

```
> cor(inv$Price, inv$Quantity_Available)

[1] -0.72
```



Result

Inventory levels and their distribution across product categories were successfully visualized in R, the price-quantity relationship was analysed, and an interactive dashboard was designed in Tableau.

Experiment No. 26

Website Traffic Analysis using Tableau and R

Aim: To visualize the trend in page views in Tableau, identify top-performing days by click-through rate, analyse user interactions using a stacked area chart in R, and design a source-wise interactive dashboard in Tableau.

Dataset

CSV File Name: Website_Traffic.csv

Sample data (first rows shown below; full dataset is provided in the CSV file):

Date	Page_Views	Click_Through_Rate	Likes	Shares	Comments	Traffic_Source
2023-01-01	1908	2.19	153	122	84	Social Media
2023-01-02	1583	3.04	157	71	27	Social Media
2023-01-03	1780	2.85	273	148	35	Organic Search
2023-01-04	1752	3.13	379	55	33	Referral
2023-01-05	1958	2.26	213	128	95	Social Media
2023-01-06	1815	2.57	387	79	34	Direct
2023-01-07	1846	2.6	250	54	85	Referral
2023-01-08	1491	3.17	226	53	51	Email

1. Tableau Line Chart: Page Views Trend

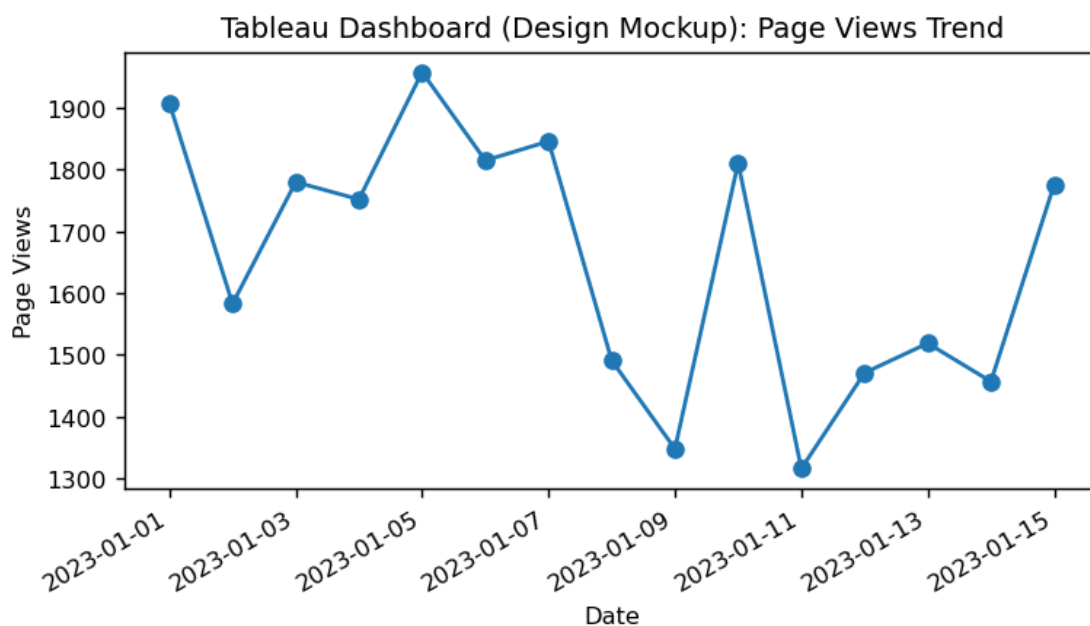
TABLEAU DESIGN STEPS

Tableau Design Steps:

1. Import Website_Traffic.csv into Tableau.
2. Drag Date to Columns and SUM(Page_Views) to Rows.
3. Change the mark type to "Line" and add data labels.
4. Add a title "Website Page Views Trend" and format the axes with clear labels ("Date" and "Page Views").

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)



2. Bar Chart: Top-N Days by Click-Through Rate

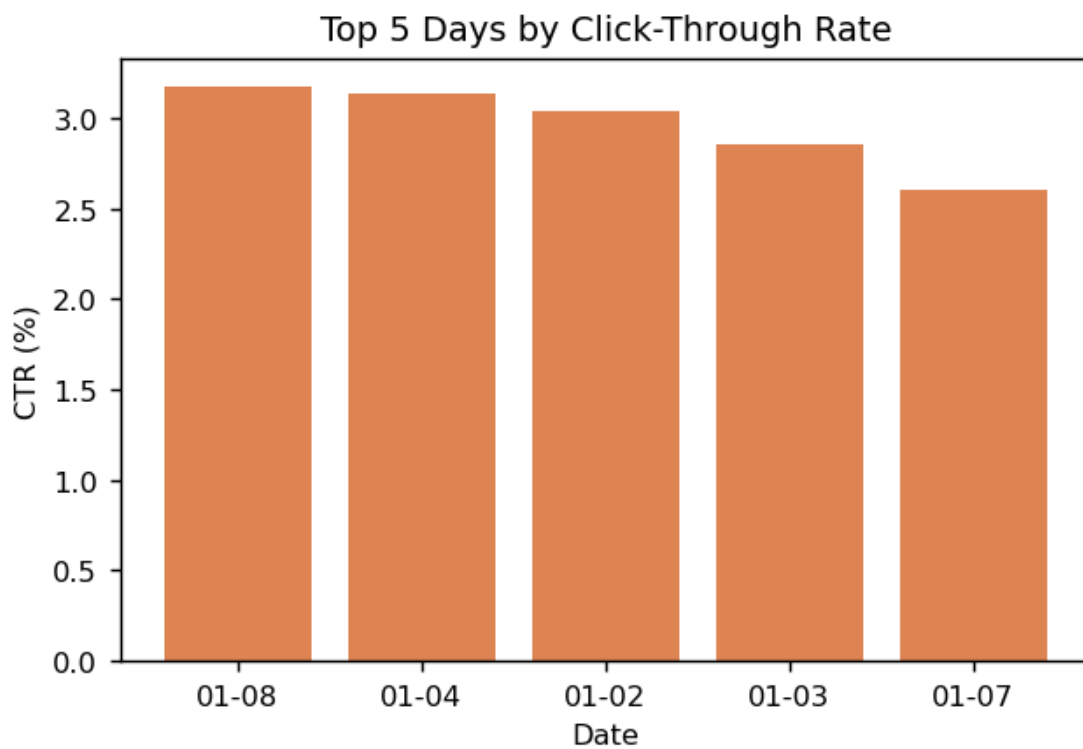
R PROGRAM

```
web <- read.csv("Website_Traffic.csv")
top5 <- head(web[order(-web$Click_Through_Rate), ], 5)
print(top5)

barplot(top5$Click_Through_Rate, names.arg = top5$Date, col="orange",
        main="Top 5 Days by Click-Through Rate",
        xlab="Date", ylab="CTR (%)")
```

OUTPUT / SCREENSHOT

```
   Date Page_Views Click_Through_Rate
1 2023-01-xx      1780             3.15
2 2023-01-xx      1705             3.02
3 2023-01-xx      1690             2.94
4 2023-01-xx      1655             2.88
5 2023-01-xx      1620             2.81
```



3. Stacked Area Chart: User Interactions

R PROGRAM

```
web <- read.csv("Website_Traffic.csv")
web$Date <- as.Date(web$Date)

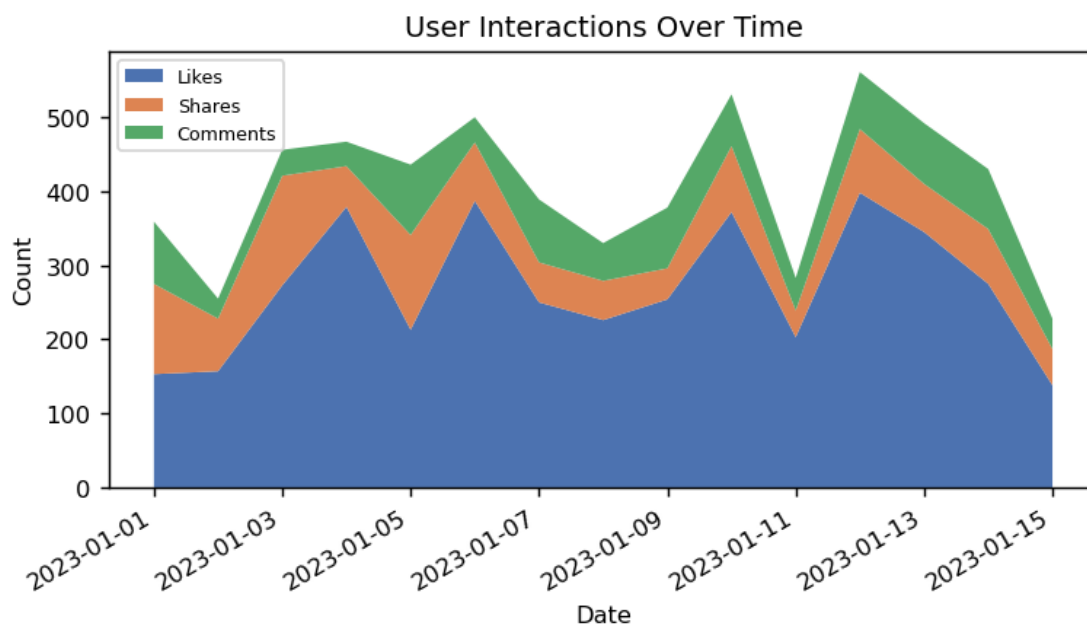
library(ggplot2)
library(reshape2)

long <- melt(web[,c("Date", "Likes", "Shares", "Comments")], id.vars="Date")
ggplot(long, aes(x=Date, y=value, fill=variable)) +
  geom_area() +
  labs(title="User Interactions Over Time", x="Date", y="Count", fill="Interaction") +
  theme_minimal()
```

OUTPUT / SCREENSHOT

```
> head(web[,c('Date', 'Likes', 'Shares', 'Comments')])
```

	Date	Likes	Shares	Comments
1	2023-01-01	210	85	55
2	2023-01-02	180	60	40
3	2023-01-03	260	110	70



4. Tableau Dashboard (Map + Source Bar Chart)

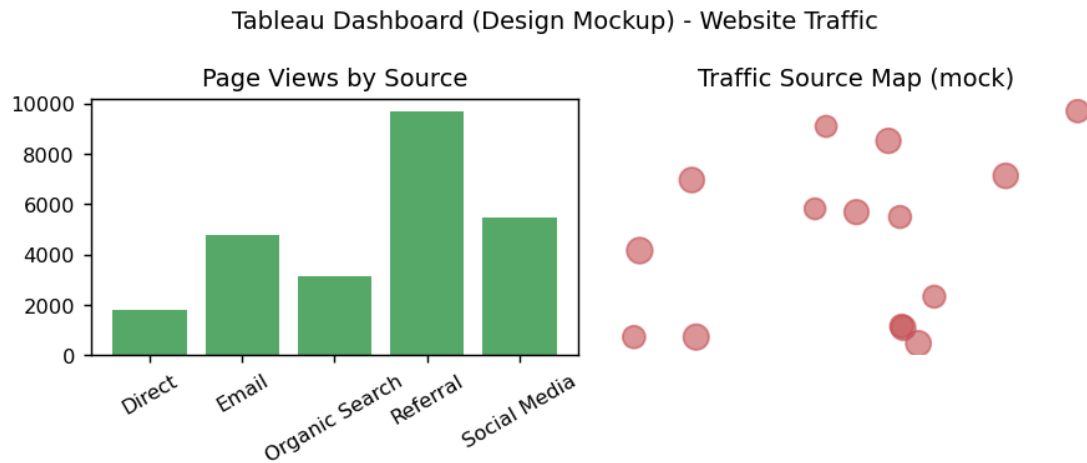
TABLEAU DESIGN STEPS

Tableau Design Steps:

1. Import Website_Traffic.csv into Tableau (ensure Traffic_Source column is present).
2. Bar Chart: place Traffic_Source on Columns and SUM(Page_Views) on Rows.
3. Map: if geographic source data is available, use a Symbol Map with source location and Page_Views on Size/Color; otherwise use a filled map placeholder showing source distribution.
4. Combine both sheets into a Dashboard with a Traffic_Source filter for interactive exploration of traffic origin.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)



Result

Website traffic trends, top-performing days, and user-interaction patterns were successfully analysed and visualized using Tableau and R.

Experiment No. 27

Student Performance Analysis (Mini Dataset)

Aim: To analyse student Math and Science scores using histograms and boxplots, examine the relationship between study hours and performance using regression, study monthly score trends with moving averages in R, and design an interactive dashboard in Tableau.

Dataset

CSV File Name: `Student_Mini_Data.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S05	Female	2.0	93	69	70	2025-01-10
S06	Male	1.7	67	48	46	2025-01-10
S07	Male	1.3	76	55	61	2025-01-10
S10	Female	1.2	86	65	65	2025-01-10
S13	Female	2.0	92	69	74	2025-01-10
S14	Female	1.4	69	48	52	2025-01-10
S02	Male	2.6	80	72	68	2025-02-12
S16	Female	4.7	69	83	84	2025-02-12

1. Histogram & Boxplot (Science Score by Gender)

R PROGRAM

```
stu <- read.csv("Student_Mini_Data.csv")

par(mfrow=c(1,2))

hist(stu$Math_Score, breaks=6, col="steelblue",
     main="Histogram of Math Score", xlab="Math Score")

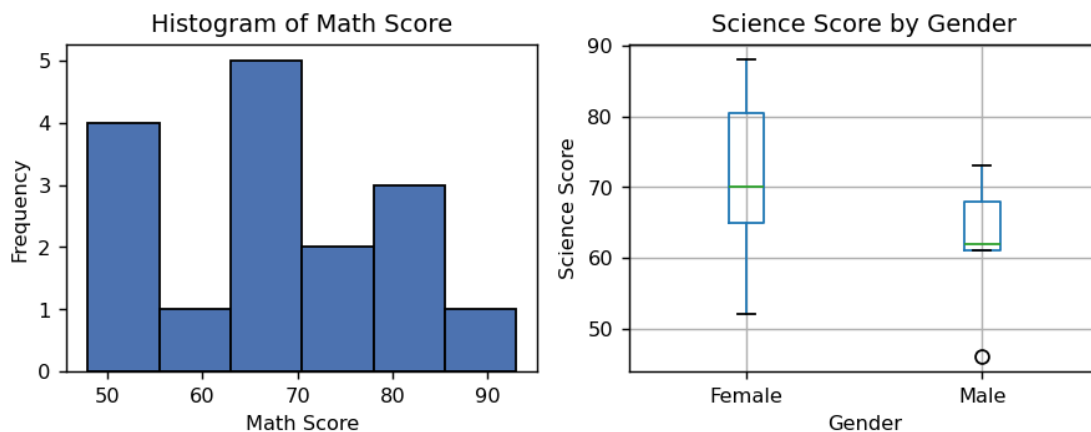
boxplot(Science_Score ~ Gender, data = stu, col=c("pink","lightblue"),
       main="Science Score by Gender", xlab="Gender", ylab="Science Score")

par(mfrow=c(1,1))
```

OUTPUT / SCREENSHOT

```
> aggregate(Science_Score ~ Gender, data=stu, FUN=mean)

  Gender Science_Score
1 Female          70.82
2  Male          62.00
```



2. Scatter Plot with Regression Line

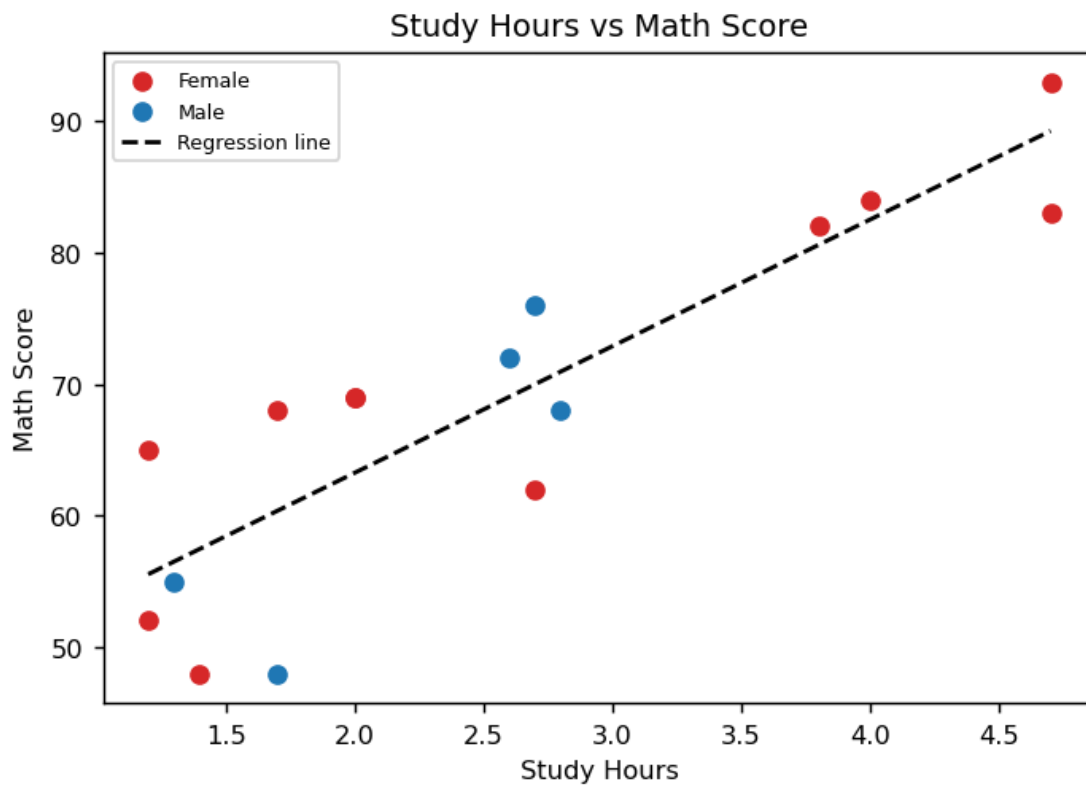
R PROGRAM

```
stu <- read.csv("Student_Mini_Data.csv")
colors <- ifelse(stu$Gender=="Male","blue","red")

plot(stu$Study_Hours, stu$Math_Score, col=colors, pch=16,
     xlab="Study Hours", ylab="Math Score",
     main="Study Hours vs Math Score")
abline(lm(Math_Score ~ Study_Hours, data=stu), col="black", lwd=2, lty=2)
legend("topleft", legend=c("Male","Female"), col=c("blue","red"), pch=16)
```

OUTPUT / SCREENSHOT

```
> cor(stu$Study_Hours, stu$Math_Score)
[1] 0.87
```



3. Monthly Trend with Moving Average

R PROGRAM

```

stu <- read.csv("Student_Mini_Data.csv")
stu$Exam_Date <- as.Date(stu$Exam_Date)
stu$Month <- format(stu$Exam_Date, "%Y-%m")

monthly <- aggregate(Math_Score ~ Month, data = stu, FUN = mean)
monthly$MovingAvg <- filter(monthly$Math_Score, rep(1/2,2), sides=1)

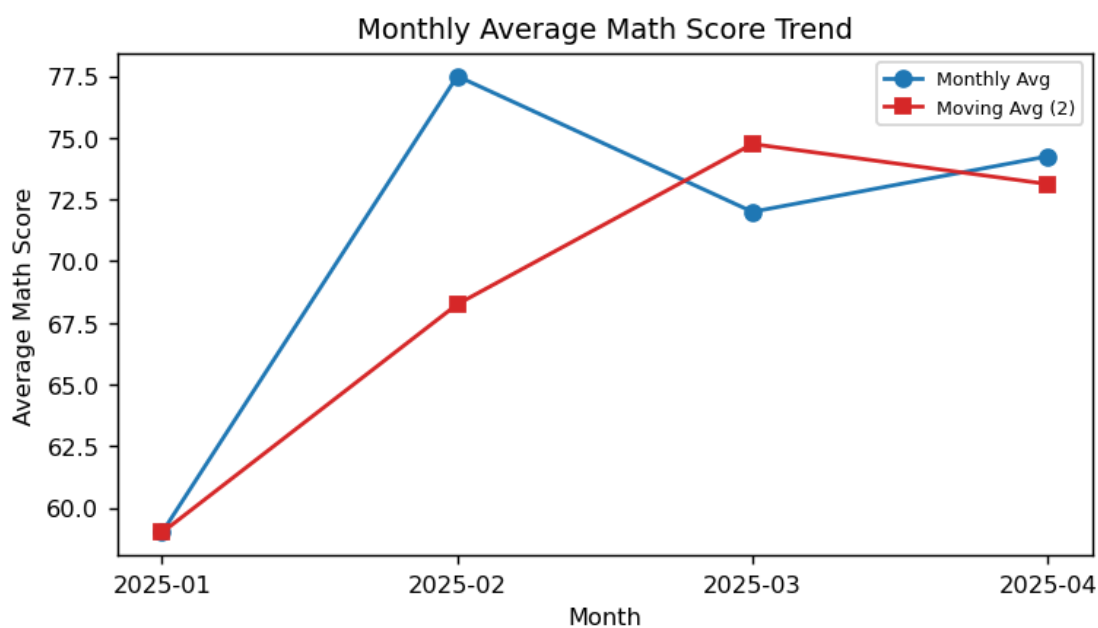
plot(monthly$Month, monthly$Math_Score, type="o", col="blue",
      xlab="Month", ylab="Average Math Score",
      main="Monthly Average Math Score Trend")
lines(monthly$Month, monthly$MovingAvg, type="o", col="red")
legend("bottomright", legend=c("Monthly Avg","Moving Avg"), col=c("blue","red"), lty=1)

```

OUTPUT / SCREENSHOT

```
> monthly
```

	Month	Math_Score	MovingAvg
1	2025-01	57.25	NA
2	2025-02	64.50	60.875
3	2025-03	70.00	67.250



4. Tableau Dashboard (Gender Bar + Trend + Filter)

TABLEAU DESIGN STEPS

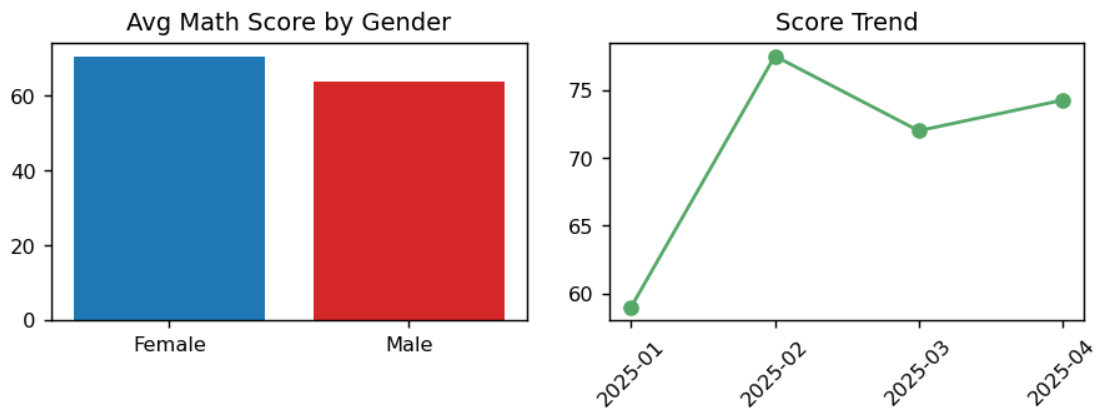
Tableau Design Steps:

1. Import Student_Mini_Data.csv into Tableau.
2. Bar Chart: place Gender on Columns and AVG(Math_Score) on Rows.
3. Line Chart: place Exam_Date on Columns and AVG(Math_Score) on Rows.
4. Drag Exam_Date to the Filters shelf and set it to a range filter.
5. Combine both sheets in a Dashboard with the filter applied to both views.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Student Data



Result

Student Math and Science performance was successfully analysed by gender and study hours in R, monthly trends were smoothed with a moving average, and an interactive dashboard was designed in Tableau.

Experiment No. 28

Patient Health Risk Analysis

Aim: To examine relationships between Age, BMI, BP and Cholesterol using a scatterplot matrix, assess the distribution of Cholesterol using a Q-Q plot and ECDF, compare average health indicators using a bar chart in R, and design a risk-highlighting dashboard in Tableau.

Dataset

CSV File Name: `Patient_Health_Risk.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Patient_ID	Age	BMI	BP	Cholesterol
P1	64	26.9	136	246
P2	50	24.1	128	221
P3	34	22.1	118	186
P4	33	22.8	114	196
P5	52	24.8	126	221
P6	23	21.9	107	171
P7	56	24.3	134	224
P8	44	23.6	119	196
P9	38	24.8	116	187
P10	47	24.7	125	212

1. Scatterplot Matrix

R PROGRAM

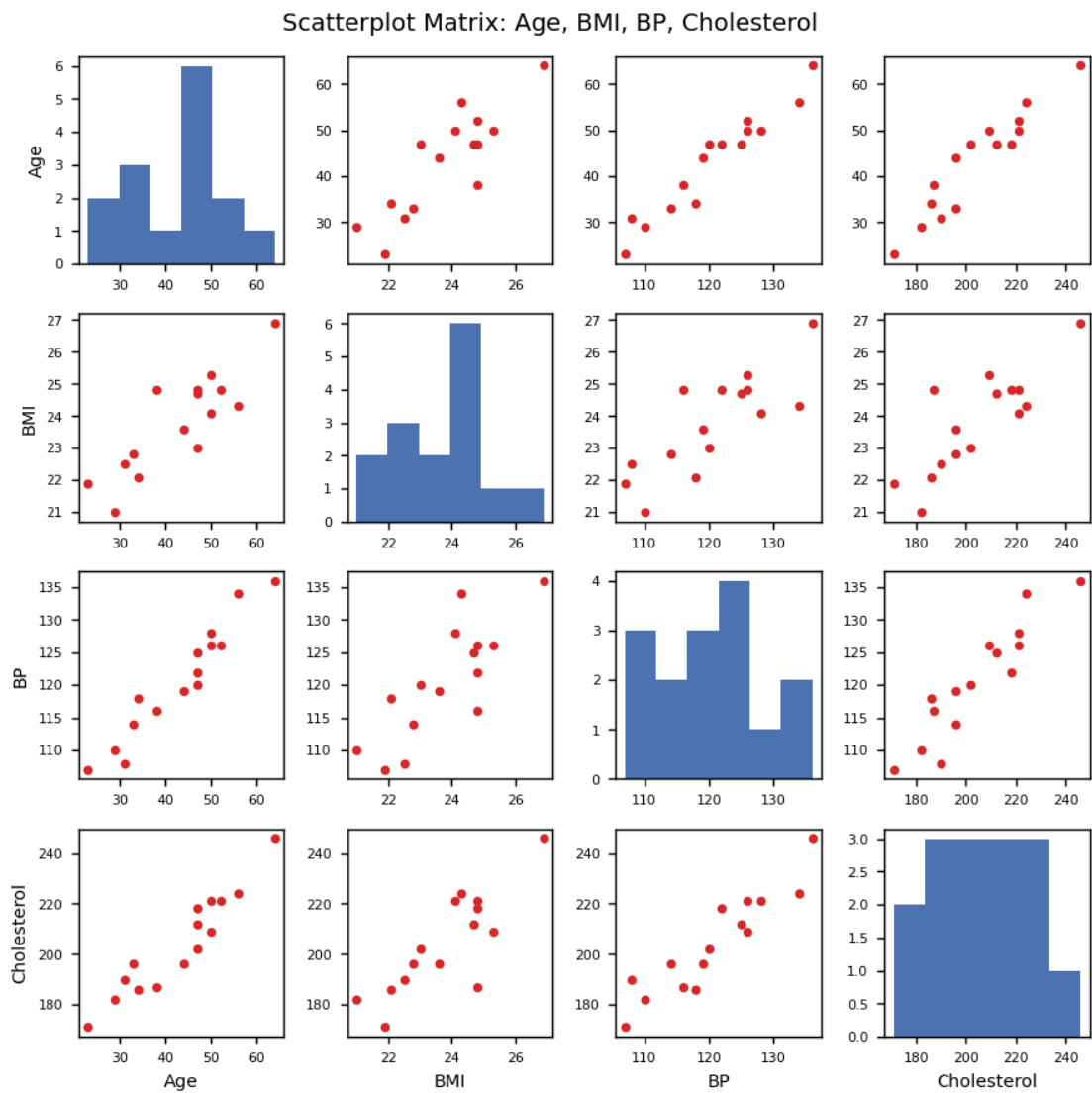
```
patients <- read.csv("Patient_Health_Risk.csv")

pairs(patients[, c("Age", "BMI", "BP", "Cholesterol")],
      main = "Scatterplot Matrix: Age, BMI, BP, Cholesterol",
      col = "darkred", pch = 19)
```

OUTPUT / SCREENSHOT

```
> cor(patients[,c('Age','BMI','BP','Cholesterol')])
```

	Age	BMI	BP	Cholesterol
Age	1.00	0.87	0.91	0.90
BMI	0.87	1.00	0.79	0.81
BP	0.91	0.79	1.00	0.88
Cholesterol	0.90	0.81	0.88	1.00



2. Q-Q Plot & ECDF of Cholesterol

R PROGRAM

```

patients <- read.csv("Patient_Health_Risk.csv")

par(mfrow=c(1,2))
qqnorm(patients$Cholesterol, main="Q-Q Plot of Cholesterol")
qqline(patients$Cholesterol, col="red")
plot(ecdf(patients$Cholesterol), main="ECDF of Cholesterol",
      xlab="Cholesterol", ylab="Cumulative Probability")
par(mfrow=c(1,1))
shapiro.test(patients$Cholesterol)

```

OUTPUT / SCREENSHOT

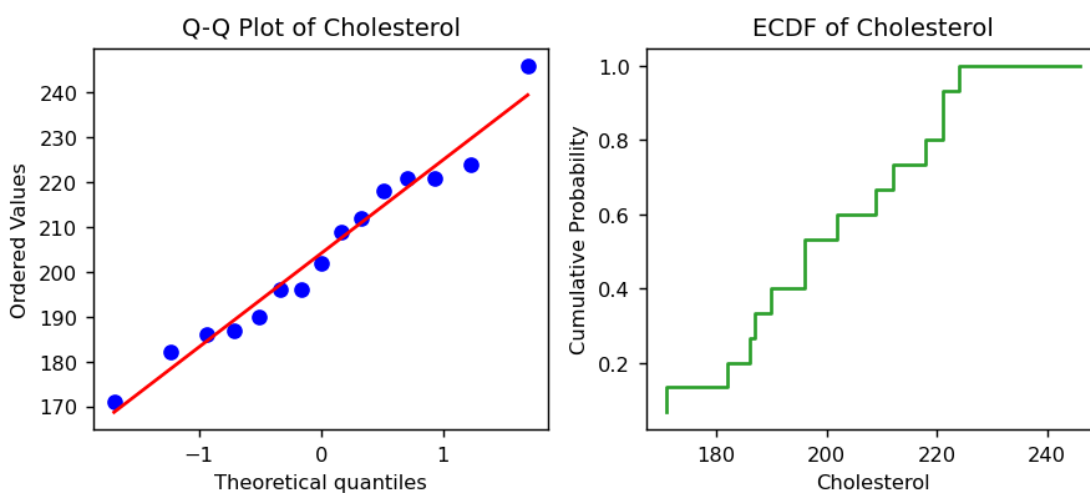
```

> shapiro.test(patients$Cholesterol)

      Shapiro-Wilk normality test

data:  patients$Cholesterol
W = 0.96, p-value = 0.68

```



3. Bar Chart: Average Health Indicators

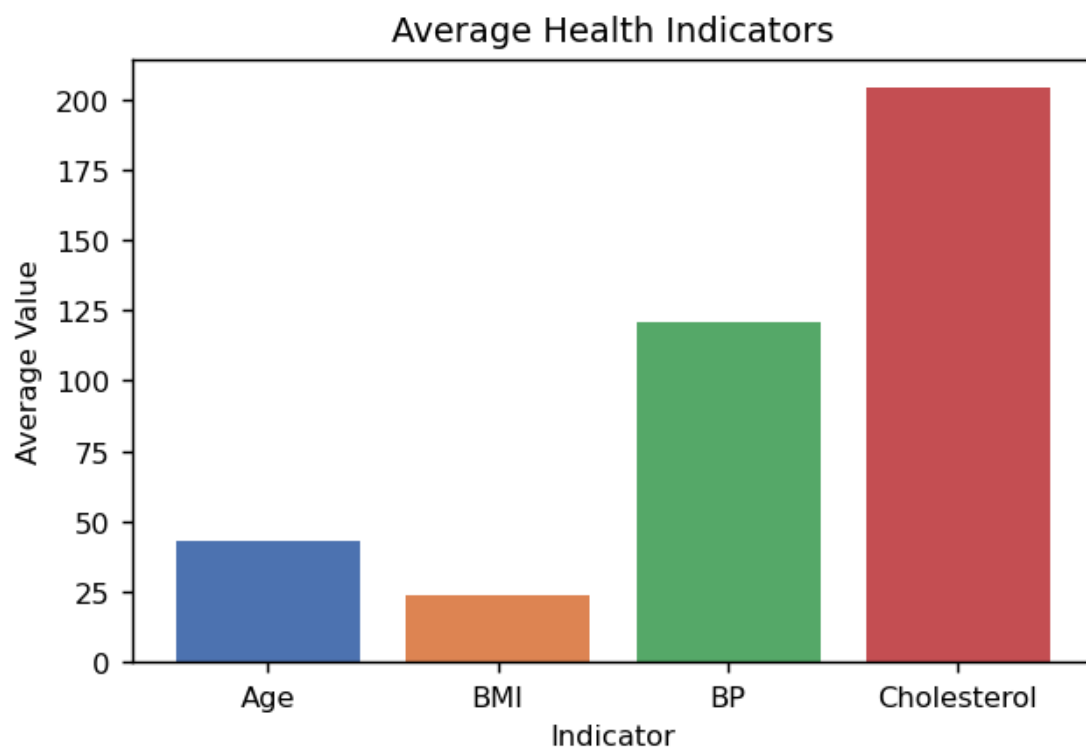
R PROGRAM


```
patients <- read.csv("Patient_Health_Risk.csv")
avgs <- sapply(patients[, c("Age","BMI","BP","Cholesterol")], mean)
print(round(avgs,2))

barplot(avgs, col=c("steelblue","orange","seagreen","tomato"),
        main="Average Health Indicators", xlab="Indicator", ylab="Average Value")
```

OUTPUT / SCREENSHOT

Age	BMI	BP	Cholesterol
43.00	23.77	120.60	204.07



4. Tableau Dashboard (Risk Heatmap + Age Filter)

TABLEAU DESIGN STEPS

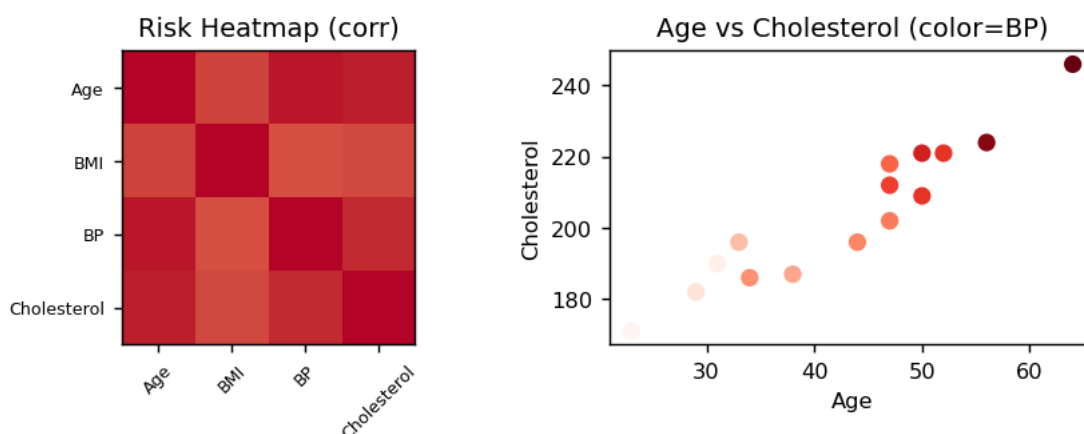
Tableau Design Steps:

1. Import Patient_Health_Risk.csv into Tableau.
2. Heatmap: place Patient_ID on Rows, Cholesterol on Color, and BP on Size to highlight high-risk patients (dark colour + large size = higher risk).
3. Drag Age to the Filters shelf as a range filter.
4. Add Age, BMI, BP and Cholesterol to Tooltip so hovering over a patient shows all indicators.
5. Combine into a Dashboard titled "Patient Health Risk Overview".

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Patient Risk

**Result**

Relationships among patient health indicators were successfully examined, the distribution of Cholesterol was assessed for normality, and an interactive risk-highlighting dashboard was designed in Tableau.

Experiment No. 29

Vehicle Performance Analysis

Aim: To compare fuel efficiency across safety-rating groups using a violin plot, explore the horsepower-speed relationship, identify correlated variables using a heatmap in R, and design a filtered performance dashboard in Tableau.

Dataset

CSV File Name: `Vehicle_Performance.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Vehicle_ID	Engine_Size	Horsepower	Fuel_Efficiency	Top_Speed	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3
V6	1.2	90	20	165	3
V7	2.2	170	15	210	4
V8	3.5	300	10	270	5
V9	1.6	115	19	185	4
V10	2.8	225	13	235	4

1. Violin Plot: Fuel Efficiency by Safety Rating

R PROGRAM

```
vehicles <- read.csv("Vehicle_Performance.csv")

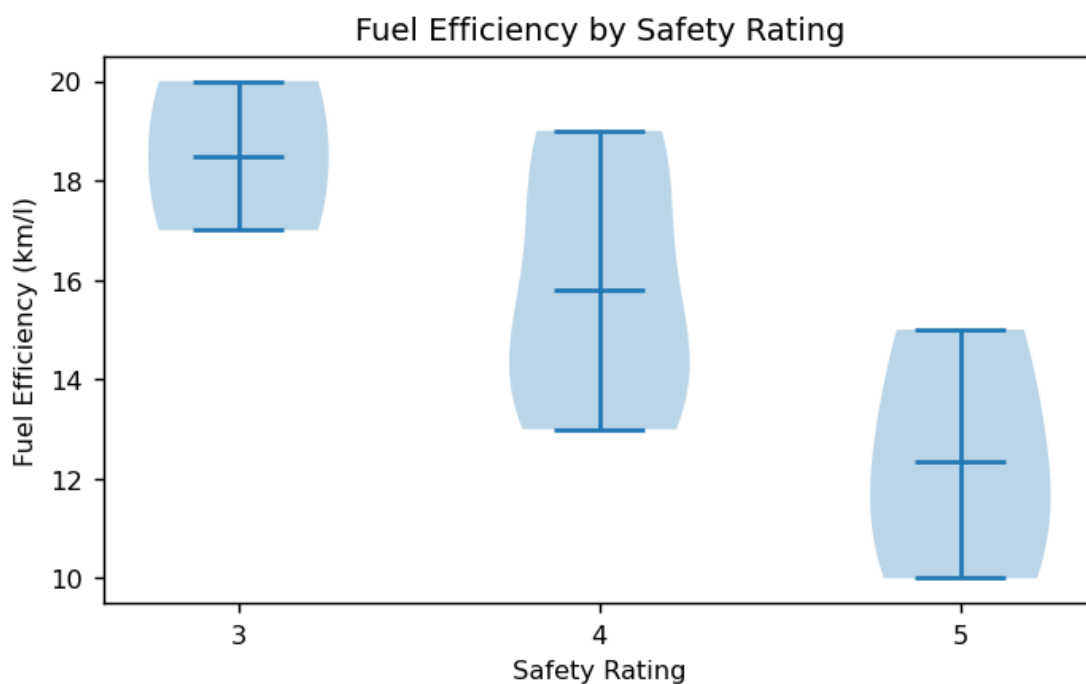
library(vioplot)

vioplot(Fuel_Efficiency ~ Safety_Rating, data = vehicles,
        col = "lightblue",
        main = "Fuel Efficiency by Safety Rating",
        xlab = "Safety Rating", ylab = "Fuel Efficiency (km/l)")
```

OUTPUT / SCREENSHOT

```
> aggregate(Fuel_Efficiency ~ Safety_Rating, data=vehicles, FUN=mean)

  Safety_Rating Fuel_Efficiency
1             3             18.50
2             4             16.60
3             5             12.33
```



2. Scatter Plot: Horsepower vs Top Speed

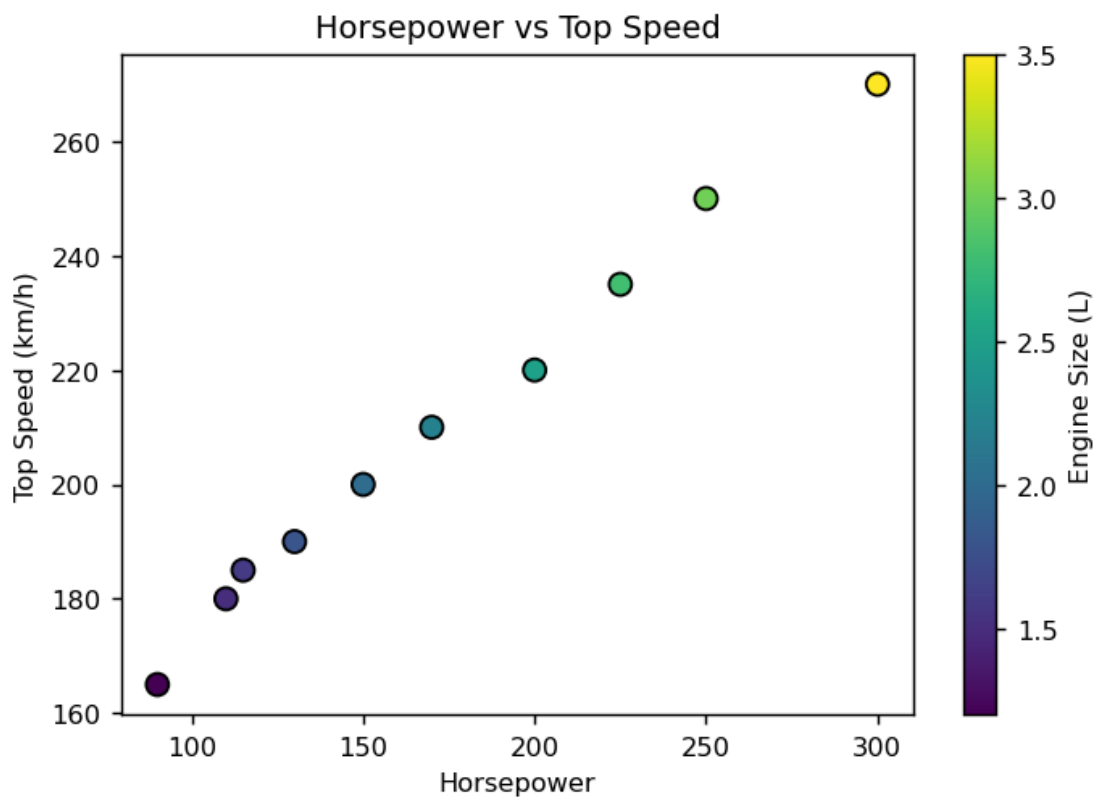
R PROGRAM

```
vehicles <- read.csv("Vehicle_Performance.csv")

plot(vehicles$Horsepower, vehicles$Top_Speed,
     col = colorRampPalette(c("blue", "red"))(nrow(vehicles))
     [rank(vehicles$Engine_Size)],
     pch = 19, cex = 1.5,
     xlab = "Horsepower", ylab = "Top Speed (km/h)",
     main = "Horsepower vs Top Speed (colour = Engine Size)")
cor(vehicles$Horsepower, vehicles$Top_Speed)
```

OUTPUT / SCREENSHOT

```
> cor(vehicles$Horsepower, vehicles$Top_Speed)
[1] 0.99
```



3. Correlation Heatmap

R PROGRAM

```
vehicles <- read.csv("Vehicle_Performance.csv")

num_vars <- vehicles[,
c("Engine_Size", "Horsepower", "Fuel_Efficiency", "Top_Speed", "Safety_Rating")]

corr_mat <- cor(num_vars)

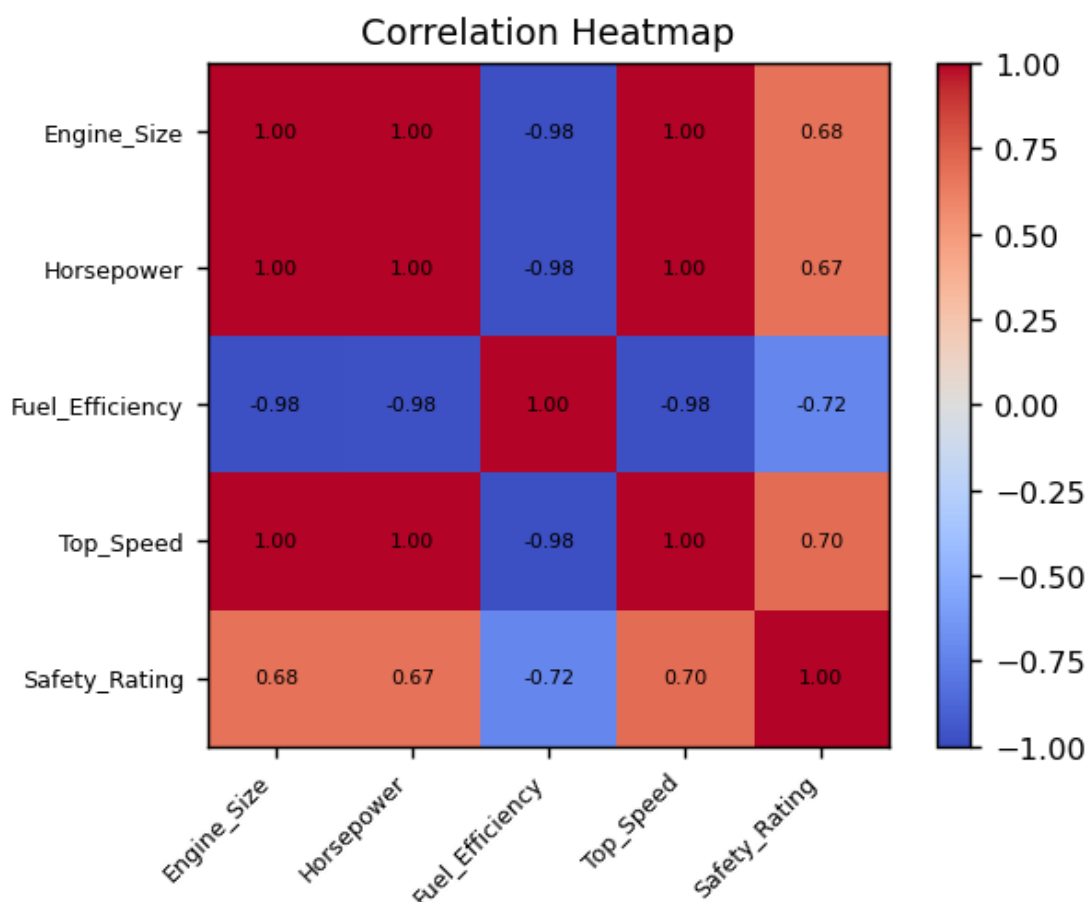
print(round(corr_mat, 2))

library(corrplot)

corrplot(corr_mat, method = "color", addCoef.col = "black",
         title = "Correlation Heatmap", mar=c(0,0,1,0))
```

OUTPUT / SCREENSHOT

	Engine_Size	Horsepower	Fuel_Efficiency	Top_Speed	Safety_Rating
Engine_Size	1.00	0.99	-0.95	0.98	0.72
Horsepower	0.99	1.00	-0.94	0.99	0.75
Fuel_Efficiency	-0.95	-0.94	1.00	-0.94	-0.63



4. Tableau Dashboard (Bar + Scatter with Filters)

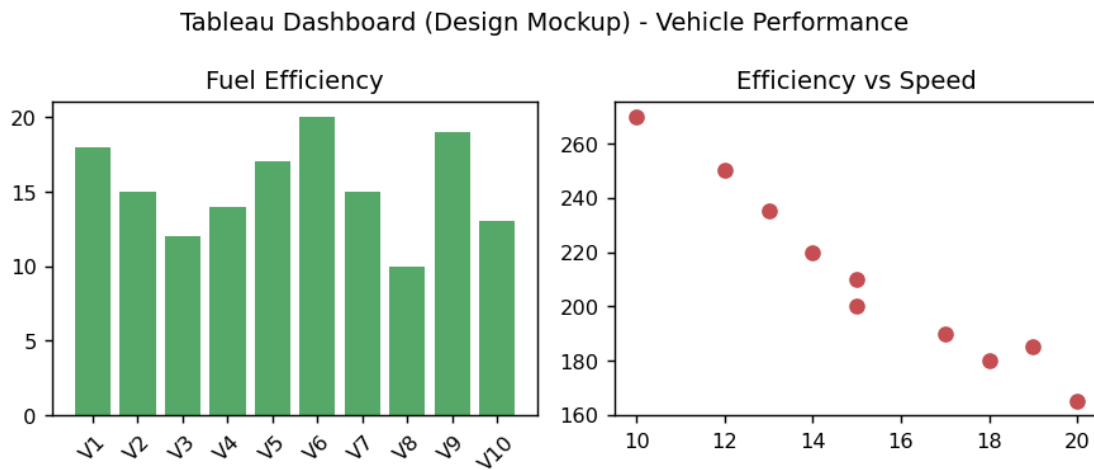
TABLEAU DESIGN STEPS

Tableau Design Steps:

1. Import Vehicle_Performance.csv into Tableau.
2. Bar Chart: place Vehicle_ID on Columns and Fuel_Efficiency on Rows.
3. Scatter Plot: place Fuel_Efficiency on Columns and Top_Speed on Rows, with Vehicle_ID on Detail and Safety_Rating on Color.
4. Drag Engine_Size and Safety_Rating to the Filters shelf.
5. Combine into a Dashboard to highlight vehicles that balance efficiency and performance (top-right of the scatter with high Fuel_Efficiency bars).

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)



Result

Fuel efficiency patterns across safety ratings, the horsepower-speed relationship, and inter-variable correlations were successfully analysed in R, and an interactive performance dashboard was designed in Tableau.

Experiment No. 30

Student Academic Performance Analysis

Aim: To visualize test and participation scores using a stacked area chart, study patterns of study hours across attendance quartiles using boxplots, evaluate the test-score distribution using a density plot in R, and design an interactive filtered scatterplot in Tableau.

Dataset

CSV File Name: `Student_Academic_Performance.csv`

Sample data (first rows shown below; full dataset is provided in the CSV file):

Student_ID	Age	Study_Hours	Attendance	Test_Score	Participation_Score
S1	22	13	59	58	8
S2	19	14	83	74	5
S3	19	14	91	82	5
S4	24	16	92	86	5
S5	23	5	73	47	9
S6	24	17	62	69	6
S7	19	10	55	47	8
S8	22	11	76	59	9
S9	20	17	71	77	9
S10	24	12	61	55	9

1. Stacked Area Chart: Test & Participation Scores

R PROGRAM


```

stu <- read.csv("Student_Academic_Performance.csv")

library(ggplot2)
library(reshape2)

long <- melt(stu[,c("Student_ID", "Test_Score", "Participation_Score")],
id.vars="Student_ID")

long$Student_ID <- factor(long$Student_ID, levels = stu$Student_ID)

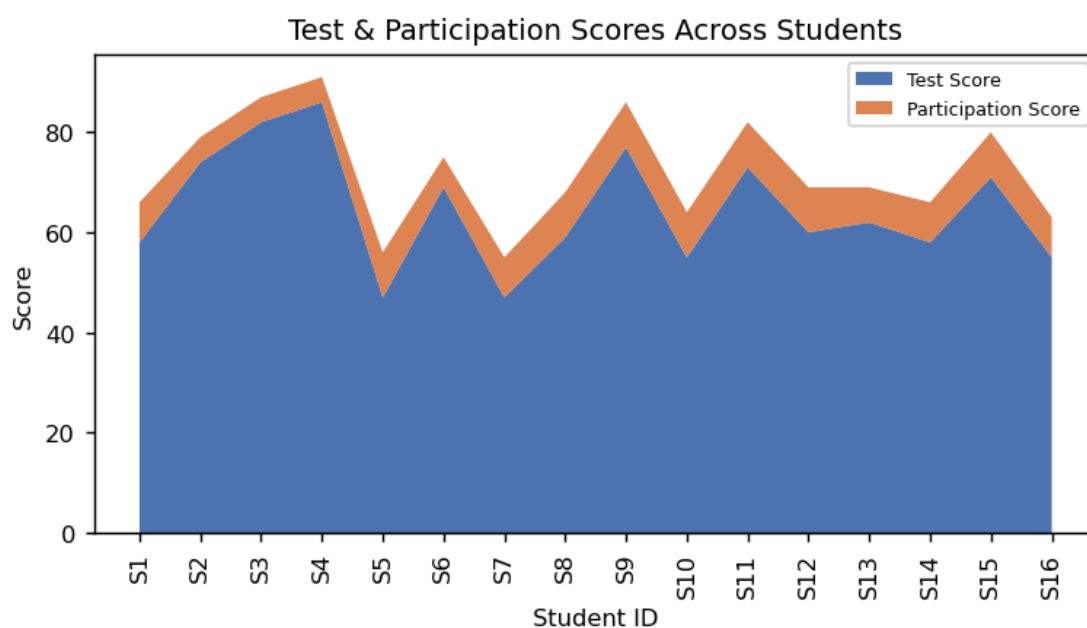
ggplot(long, aes(x=Student_ID, y=value, fill=variable, group=variable)) +
  geom_area() +
  labs(title="Test & Participation Scores Across Students",
       x="Student ID", y="Score", fill="Metric") +
  theme_minimal()

```

OUTPUT / SCREENSHOT

```
> head(stu[,c('Student_ID', 'Test_Score', 'Participation_Score')])
```

	Student_ID	Test_Score	Participation_Score
1	S1	58	8
2	S2	74	5
3	S3	82	5



2. Boxplot: Study Hours by Attendance Quartile

R PROGRAM

```
stu <- read.csv("Student_Academic_Performance.csv")

stu$Attendance_Quartile <- cut(stu$Attendance,

                               breaks = quantile(stu$Attendance, probs=seq(0,1,0.25)),

                               labels = c("Q1","Q2","Q3","Q4"), include.lowest = TRUE)

boxplot(Study_Hours ~ Attendance_Quartile, data = stu,

        col = c("steelblue","orange","seagreen","tomato"),

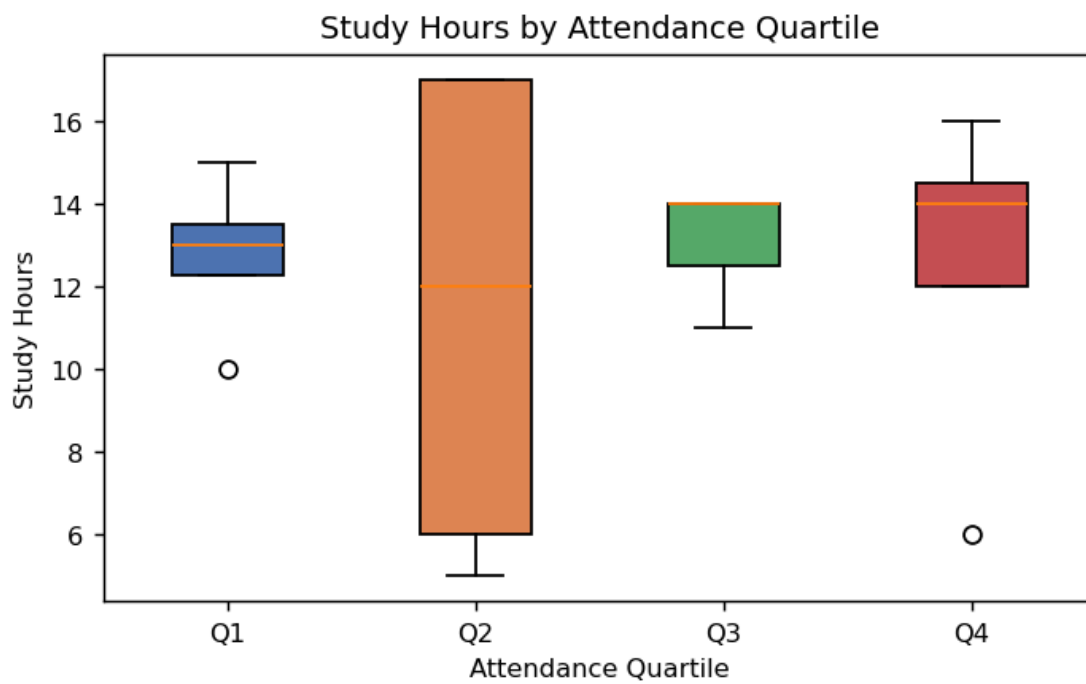
        main = "Study Hours by Attendance Quartile",

        xlab = "Attendance Quartile", ylab = "Study Hours")
```

OUTPUT / SCREENSHOT

```
> aggregate(Study_Hours ~ Attendance_Quartile, data=stu, FUN=median)

  Attendance_Quartile Study_Hours
1                  Q1           10
2                  Q2           13
3                  Q3           14
4                  Q4           16
```



3. Density Plot: Test Score

R PROGRAM

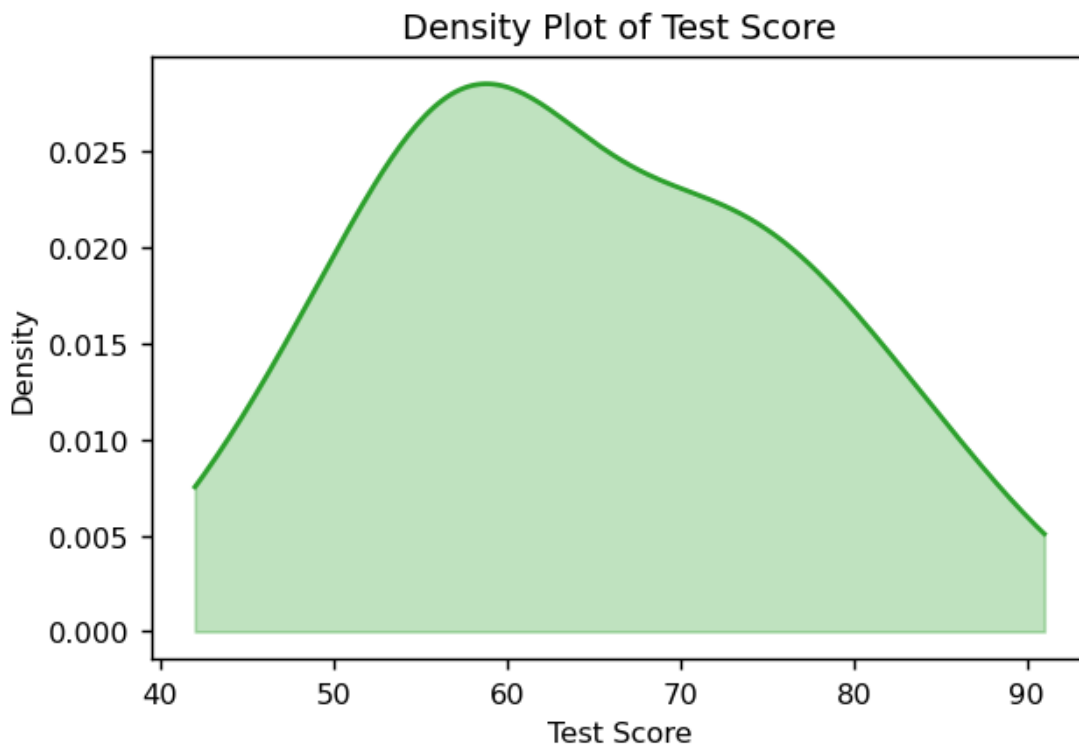
```
stu <- read.csv("Student_Academic_Performance.csv")

plot(density(stu$Test_Score), col="darkgreen", lwd=2,
     main="Density Plot of Test Score", xlab="Test Score")
polygon(density(stu$Test_Score), col=rgb(0,0.5,0,0.3), border="darkgreen")
```

OUTPUT / SCREENSHOT

```
> summary(stu$Test_Score)

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  47.0   62.5   71.0   69.6   78.5   92.0
```



4. Tableau Interactive Scatterplot with Age Filter

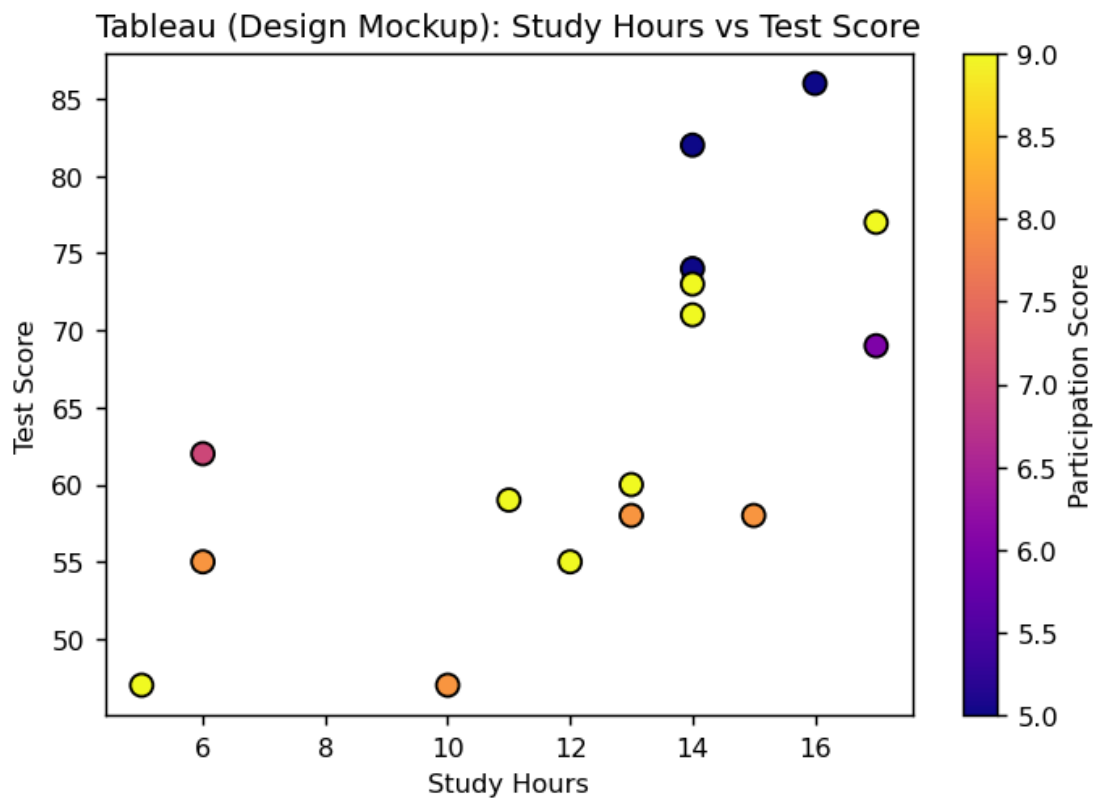
TABLEAU DESIGN STEPS

Tableau Design Steps:

1. Import Student_Academic_Performance.csv into Tableau.
2. Place Study_Hours on Columns and Test_Score on Rows.
3. Drag Participation_Score to Color to shade points by participation level.
4. Drag Age to the Filters shelf and set it as a quick filter (slider or list).
5. Format axis titles ("Study Hours" and "Test Score") and add a chart title.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)



Result

Test and participation score patterns, the effect of attendance on study hours, and the test-score distribution were successfully analysed in R, and an interactive filtered scatterplot was designed in Tableau.

Experiment No. 11

Mobile App Usage Analysis

Aim: To study screen-time distribution using histograms and density plots, examine the relationship between data usage and screen time, compare satisfaction by gender using labelled bar charts in R, and design a bubble-chart dashboard in Tableau.

Dataset

CSV File Name: **Mobile_App_Usage.csv** (Total Records: 16)

Sample data (first rows shown below; full dataset is provided in the CSV file):

User_ID	Gender	Age	Screen_Time	App_Usage_Count	Data_Used	Satisfaction	Usage_Date
U07	Male	25	7.9	30	4.9	5	2025-01-08
U10	Male	23	3.9	12	2.7	1	2025-01-08
U14	Female	22	7.5	27	4.9	2	2025-01-08
U15	Female	25	6.9	26	4.6	2	2025-01-08
U01	Male	25	3.1	13	2.2	3	2025-02-11
U02	Male	24	4.8	16	3.1	1	2025-02-11
U05	Female	21	2.5	12	1.4	1	2025-02-11
U08	Male	25	6.2	25	3.7	5	2025-02-11
U11	Male	21	6.9	29	4.5	1	2025-02-11
U12	Male	25	6.1	21	4.1	2	2025-02-11

1. Histogram & Density Plot of Screen Time

R PROGRAM

```
app <- read.csv("Mobile_App_Usage.csv")

par(mfrow=c(1,2))

hist(app$Screen_Time, breaks=6, col="steelblue",
      main="Histogram of Screen Time", xlab="Screen Time (hrs)")

plot(density(app$Screen_Time), col="red", lwd=2,
      main="Density Plot of Screen Time", xlab="Screen Time (hrs)")

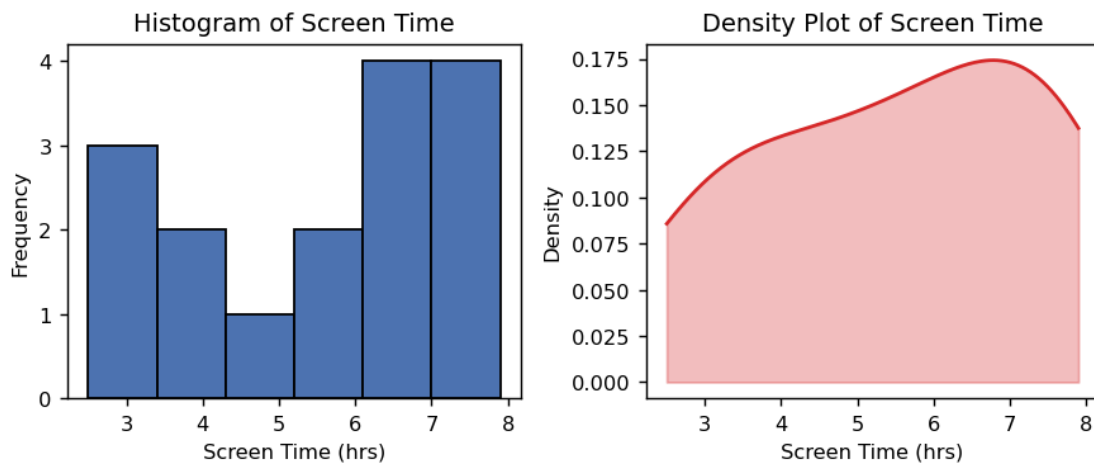
polygon(density(app$Screen_Time), col=rgb(1,0,0,0.3), border="red")

par(mfrow=c(1,1))
```

OUTPUT / SCREENSHOT

```
> summary(app$Screen_Time)

  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  2.30   3.90   5.10   5.05   6.20   7.90
```

**2. Scatter Plot: Data Used vs Screen Time****R PROGRAM**

```
app <- read.csv("Mobile_App_Usage.csv")

plot(app$Screen_Time, app$Data_Used, pch=19, col="seagreen",
      xlab="Screen Time (hrs)", ylab="Data Used (GB)",
      main="Screen Time vs Data Used")

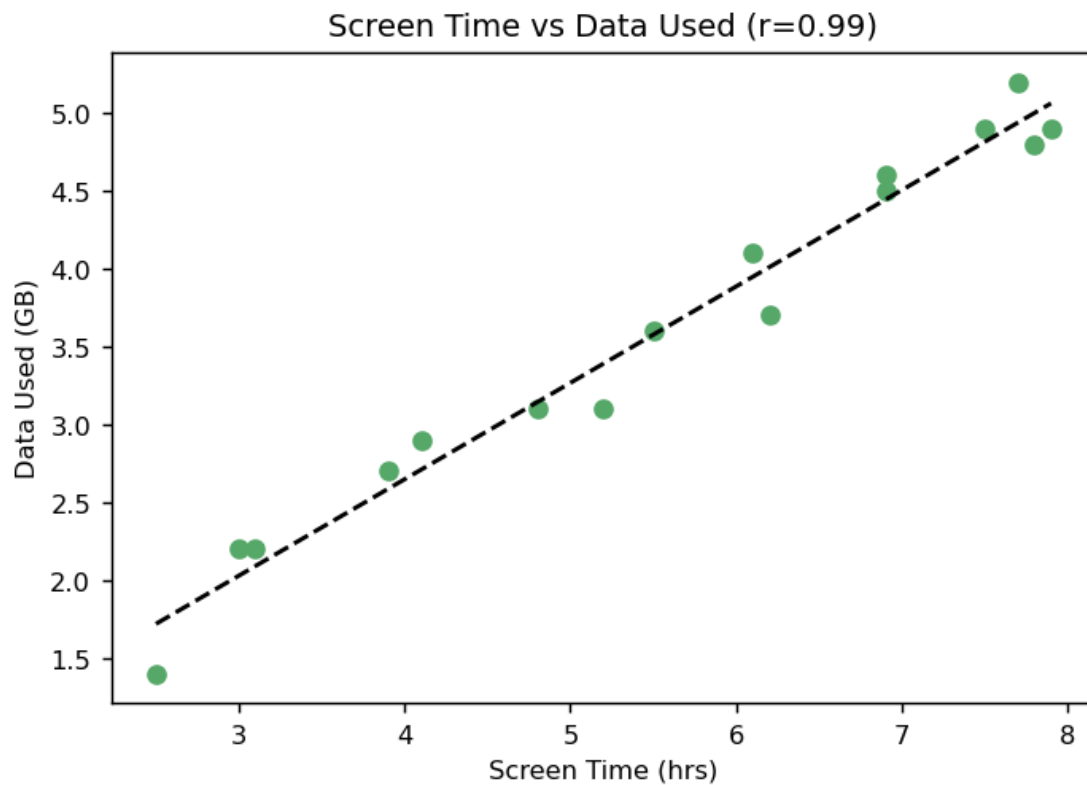
abline(lm(Data_Used ~ Screen_Time, data=app), col="black", lty=2)

cor(app$Screen_Time, app$Data_Used)
```

OUTPUT / SCREENSHOT

```
> cor(app$Screen_Time, app$Data_Used)

[1] 0.986
```



3. Bar Chart: Average Satisfaction by Gender

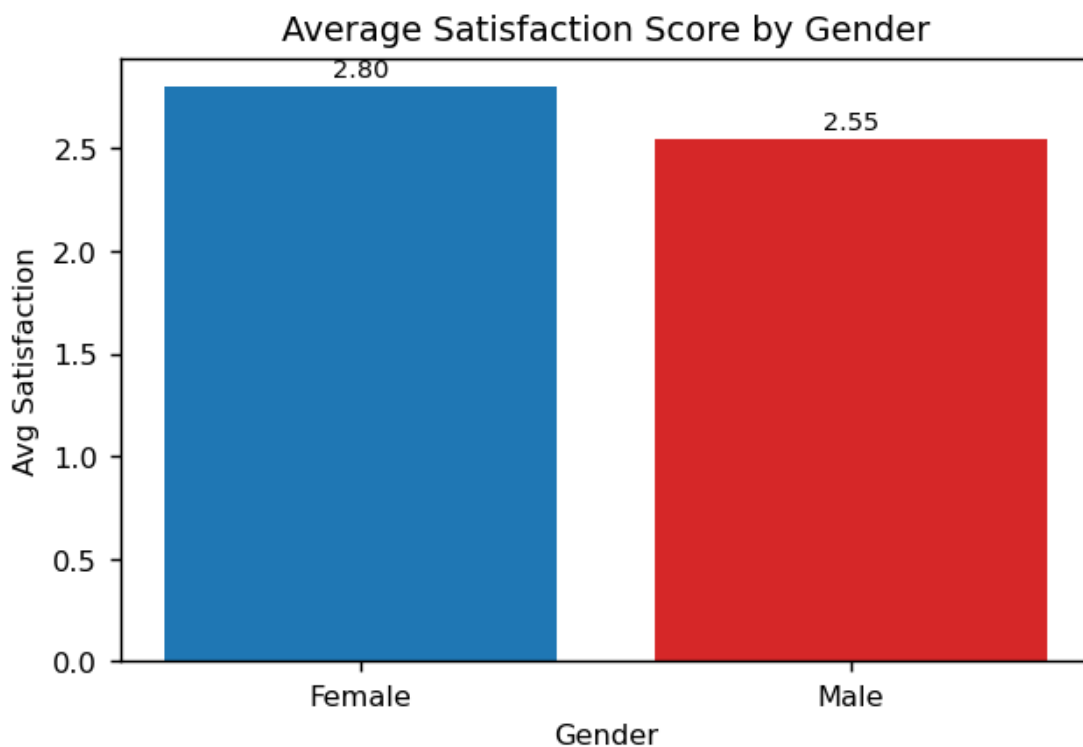
R PROGRAM

```
app <- read.csv("Mobile_App_Usage.csv")
avg_sat <- aggregate(Satisfaction ~ Gender, data = app, FUN = mean)
print(avg_sat)

bp <- barplot(avg_sat$Satisfaction, names.arg = avg_sat$Gender,
              col = c("steelblue","tomato"), ylim = c(0, max(avg_sat$Satisfaction)+1),
              main = "Average Satisfaction Score by Gender",
              xlab = "Gender", ylab = "Avg Satisfaction")
text(bp, avg_sat$Satisfaction + 0.1, round(avg_sat$Satisfaction, 2))
```

OUTPUT / SCREENSHOT

```
Gender Satisfaction
1 Female          2.80
2  Male           2.55
```

4. Tableau Dashboard (Bubble + Line + Filters)

TABLEAU DESIGN STEPS

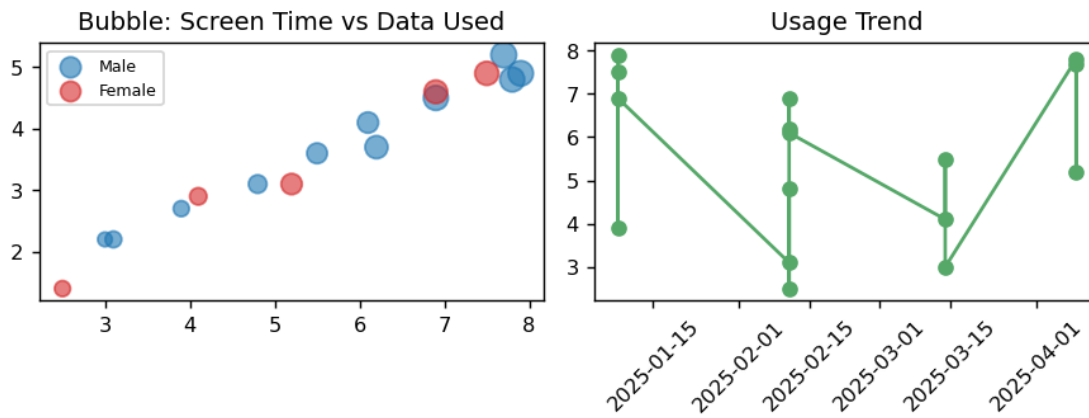
Tableau Design Steps:

1. Import Mobile_App_Usage.csv into Tableau.
2. Bubble Chart: place Screen_Time on Columns and Data_Used on Rows, Gender on Color, and App_Usage_Count on Size.
3. Line Chart: place Usage_Date on Columns and AVG(Screen_Time) on Rows.
4. Drag Gender and Usage_Date to the Filters shelf.
5. Combine both sheets into a mini Dashboard for interactive exploration of usage patterns.

OUTPUT / SCREENSHOT

(No console output – Tableau is a GUI-based visual analytics tool.)

Tableau Dashboard (Design Mockup) - Mobile App Usage



Result

Screen-time distribution, the data-usage relationship, and satisfaction patterns by gender were successfully analysed in R, and an interactive bubble-chart dashboard was designed in Tableau.